

Master Thesis

Self-Organization Patterns in LLM-based Multi-Agent Systems

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Munich, October 4th, 2026



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Abstract

A popular way to improve how language models reason is multi-agent debate, in which several agents discuss a problem over multiple rounds. As such systems move into real applications, it matters not only whether they reach good answers but also whether the way they reach them can be trusted. Recent studies raise doubts on both counts, showing that debate often does no better than a simple vote, as groups agree too early on what they already share and agents abandon correct answers to follow confident peers. These failures arise from how the discussion unfolds, yet debate is mostly judged by outcome measures such as final accuracy. Such measures cannot tell apart two debates that reach the same answer, one because the group corrected an early mistake and the other because it simply amplified the answer most of its agents started with. Studying the discussion itself calls for a theory of how interacting parts produce collective behaviour, which is exactly what complex systems theory offers. If debate behaves like other complex systems, its failures should leave recognisable patterns that explain why it goes wrong and could signal problems early. We connect this theory to multi-agent debate and introduce Self-Organization Motif Analysis (SOMA), a method that detects four such patterns in debates, from self-reinforcing positions to emerging leaders.

Applied to 30,000 debates, these patterns reveal a single mechanism, as debate mostly amplifies the answers agents hold at the start rather than producing new ones. When we change only how many agents start with the correct answer, for instance, the group ends correctly more often with each additional correct agent, while it rarely reaches a correct answer that none of its agents started with. Diverse starting answers, in turn, make it more likely that the group follows an agent who was right from the beginning. Since SOMA largely works with information that many debate systems already record, the path to an answer becomes something practitioners can measure rather than merely assume. Thus, this thesis connects complex systems theory with multi-agent debate and turns the resulting insights into concrete design guidance.

Acknowledgements

First of all, I would like to thank Ali Modarressi, who supervised this thesis at LMU Munich. He helped me turn my ideas into proper scientific work and taught me a lot about academic standards along the way. Working with Ali was a lot of fun, and I always enjoyed the friendly and relaxed atmosphere of our meetings. I would also like to thank Prof. Dr. Hinrich Schütze for making this thesis possible.

Furthermore, I am very grateful to Julius Eckhardt and Christoph Kommer at SAP. They helped me find this topic, took the time to introduce me to complex systems theory and made sure I had the language models and computing resources my experiments needed. Many of the ideas in this thesis came up in our conversations, and their feedback on my drafts helped me a lot. I really enjoyed working with both of them and am grateful for their support throughout the project.

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1 Introduction

1.1 Motivation

Large language models (LLMs) are increasingly used not as isolated models but as interacting agents that reason, argue, and decide in groups. A prominent instance of this is *multi-agent debate* (MAD), in which several LLM instances independently propose answers, exchange their reasoning over multiple rounds, and revise their positions before the group ends up with a final decision (Du et al., 2024). Across factual and mathematical reasoning tasks, this simple protocol has been reported to improve accuracy and reduce hallucination relative to a single model answering alone (Du et al., 2024, Liang et al., 2024, Chen, Saha and Bansal, 2024), and it now supports a growing family of collaborative and evaluative systems (Chan et al., 2024). And the appeal is intuitive: independent agents make partially independent errors, and deliberation offers a mechanism for the group to identify and correct them.

However, this optimistic reading is almost entirely based on *outcome* measurements. The literature evaluates multi-agent systems mostly by final task performance, treating the deliberation itself as an mysterious process that either does or does not yield a better answer. Recent critical reassessments show that this outcome-centric view is fragile: debate frequently fails to outperform single-model baselines and collapses under controlled comparison with simpler alternatives (Zhang et al., 2025, Choi et al., 2025, Bertalaníč and Fortuna, 2026). More troublingly, the failures that do occur are not random noise but exhibit structure. When two LLMs debate, both tend to grow *more* confident that they have won as the exchange proceeds, with confidence escalating away from a rational baseline even when no new supporting evidence is introduced (Prasad and Nguyen, 2025). Under distributed information, groups of LLM agents converge prematurely on the evidence they already share and fail to surface the private knowledge needed for the correct answer, so that collective accuracy collapses far below what a single agent achieves with complete information (Li et al., 2026).

What these failures have in common is that they are *dynamical*. Rising confidence, premature lock-in, and conformity behaviour are properties of how positions evolve and influence one another across rounds, and not of any single agent’s competence at a single point in time. An outcome metric, by construction, cannot see them: two systems can reach the same final accuracy through entirely different trajectories, one through genuine error correction and the other through a confident slide into a shared mistake. Understanding *when* and *why* multi-agent deliberation succeeds therefore requires a vocabulary for the interaction dynamics themselves. Miehl et al. (2025) make precisely this argument at a conceptual level, stating that agentic AI must be studied through a systems-theoretic lens because emergent group-level behaviour cannot be predicted from individual agent properties alone. However, their work remains conceptual and offers no empirical instrument for detecting such behaviour in practice.

Complex-systems theory supplies exactly such an instrument. Systems of many inter-

acting units that produce ordered collective behaviour without any central control, the defining feature of self-organization (Ashby, 1947), are known to fall into a small number of recurring dynamical regimes. These regimes have precise names and precise signatures. *Self-reinforcement*, is the positive feedback by which a state amplifies its own position (Arthur, 1989); *limit cycles* are self-sustained periodic oscillations that never settle (Strogatz, 2015); *multistability* is the coexistence of multiple stable outcomes selected by initial conditions (Feudel, 2008); and *dominance* is the concentration of influence onto a single node (DeGroot, 1974, Jia et al., 2015). Such patterns are studied as *motifs*, which are recurring structural or dynamical building blocks whose presence in a system is both detectable and diagnostic of its behaviour (Milo et al., 2002, Alon, 2007, Driscoll et al., 2024). The failure modes documented in the MAD literature read naturally as instances of this vocabulary. For instance, escalating confidence is a self-reinforcement signature, premature consensus is a fixed-point attractor reached through basin selection, and herding to a confident leader is a dominance motif. Thus, the connection has begun to be made. Ashery et al. (2025) demonstrate the spontaneous emergence of universally adopted social conventions in decentralised LLM populations, show that strong collective biases arise even when individual agents exhibit none, and find that committed adversarial minorities can impose alternative conventions on the majority. Okawa (2026) derives a formal model predicting that once conformity pressure among agents exceeds a critical threshold, even small individual biases are amplified into a shared collective bias. However, these convention-based studies all use tasks where every option is equally good. They can therefore show that a group converges, but not whether it converges on the correct answer.

This thesis addresses that gap. We construct a controlled cooperative multi-agent debate system in which homogeneous LLM agents deliberate over multiple rounds on tasks with verifiable ground truth. On top of this system, we develop a suite of detectors, each grounded in the complex-systems and opinion-dynamics literature, that identify self-reinforcement, limit cycles, multistability, and dominance in the resulting interaction logs. We run this system across a controlled grid of communication topologies and memory configurations on two complementary benchmarks: GPQA-Diamond (Rein et al., 2024), a graduate-level science question-answering benchmark where agents typically start with reasonable priors, and HiddenBench (Li et al., 2026), a hidden-profile benchmark constructed so that agents start systematically wrong and success requires genuine information pooling. The contrast between the two isolates whether observed dynamics depend on the presence of a reliable individual signal or persist even when the group must correct a shared bias. Throughout, we treat motif detection not as a competitor to accuracy but as an explanatory layer beneath it. We think of it as a way to characterise the trajectory that produces an outcome, and thereby use it to understand which design choices shape the deliberative dynamics of LLM collectives.

This explanatory layer has a practical purpose. As multi-agent systems are deployed in high risk environments, an operator needs to know not only whether a configuration tends to produce correct answers but also whether a given deliberation is proceeding soundly or in a broken way, whether the group is genuinely correcting an error or just confidently converging on a shared mistake, and whether it will terminate at all. Outcome metrics

cannot answer these questions alone. Related work has started to address this: Fan et al. (2026) train a classifier that predicts, from simple features of a query, whether debate is worth running at all. What is still missing is a tool that names which regime a deliberation is in, using the vocabulary of dynamical systems, rather than only flagging that something is wrong. Our detectors are meant to fill this gap. Because each detector is computed from ordinary interaction logs, it can serve as a candidate *diagnostic* of deliberative health. We do not deliver a validated diagnostic tool. Instead, we provide the groundwork for one, a set of measurable signatures characterised across a controlled grid, together with design principles graded by how strongly our data supports them. In one sentence, the objective of this thesis is to understand *how* a cooperative LLM debate arrives at its answer, by measuring the self-organization patterns in its trajectory, and to turn that understanding into evidence-based guidance for building such MAD systems.

1.2 Research Questions

The goal of this thesis is to establish whether the dynamical vocabulary of complex systems can be applied empirically to LLM-based multi-agent debate, and to characterise the conditions under which self-organization patterns emerge. We split this goal into three research questions, ordered by the class of evidence each requires.

RQ1 Measurement. Can self-organization motifs, specifically self-reinforcement, limit cycles, multistability, and dominance, be detected in cooperative LLM-based multi-agent debate using detectors grounded in the complex-systems and opinion-dynamics literature? And which of these motifs actually occur, at what frequency, and with what structure?

RQ2 Characterisation. What determines the presence and strength of each motif, and what do they predict about system outcomes? We examine two structural axes, communication topology and memory window, together with the information regime of the task and the composition of the opening votes, and ask what each motif implies for accuracy, convergence speed, and worst-case runtime.

RQ3 Causation. Is the composition of the opening votes causally responsible for the outcome? Can the system generate a correct answer that no agent initially held, or does deliberation function primarily as an amplifier of the initial vote distribution?

Throughout, every detector is tested against a null model that removes the pattern we look for and keeps everything else. This matters because our setup is small. With only four agents and three or four voting options, votes repeat and influence concentrates by chance alone. Thus, the null models show what chance looks like, so a motif with a very low frequency is still a result, not a failure. We report the motifs we do not find as well as those we do.

1.3 Contributions & Structure

This thesis makes three main contributions.

- **soma, an instrument for measuring debate dynamics.** A two-phase debate protocol that logs private beliefs and confidence scores separately from public messages (Section 4), together with four motif detectors, each grounded in the complex-systems or opinion-dynamics literature and tested against its own null model (Section 5).
- **A repeated-run characterisation at scale.** 30,000 debates across twelve configurations and two contrasting benchmarks, with $R = 50$ repetitions per task and configuration, which makes across-repetition attractor statistics measurable in the first place (Section 6).
- **A controlled intervention on the opening votes.** A seeded-initialisation experiment and an adversarial probe that manipulate only the composition of the opening votes while leaving the debate itself unchanged (Section 6).

All four motifs describe a single mechanism. The debate has positive feedback but no negative feedback, so it acts as a selective but lossy amplifier of its opening votes. It strengthens what the agents already hold, rarely produces an answer none of them started with, and can destroy a correct majority. From this we derive evidence-graded design guidance, mapping each recommendation to an observable that can be computed from ordinary interaction logs (Section 7).

Structure of this thesis. The thesis follows the order of the three research questions. *Background* (Section 2) introduces the vocabulary of self-organization and dynamical motifs and fixes what we mean by an LLM agent and MAD; *Related Work* (Section 3) surveys how multi-agent systems are currently analysed and locates the gap this thesis addresses. The instrument is then built in two steps. Firstly, *System and Experimental Setup* (Section 4) specifies the debate protocol, the two contrasting benchmarks, and the experimental grid, and secondly, *Detection Methodology* (Section 5) defines the four SOMA detectors together with the null model each is tested against. *Results* (Section 6) then applies the instrument, reporting accuracy and efficiency, the four motifs in turn, the emergent influence topology, and the seeded-initialisation experiment. *Discussion* (Section 7) continues by interpreting the motifs as one amplifier mechanism, traces it to the absence of negative feedback, and derives evidence-based design principles. It also states the limits of our claims and the future work they imply. *Conclusion* (Section 8) closes by answering the three research questions in turn and showing how the detectors, the characterisation, and the causal test support each other.

2 Background

This section provides the vocabulary needed to follow the rest of the thesis. Section 2.1 introduces complex systems, self-organization, and dynamical systems as the formal language for describing collective behaviour. Section 2.2 introduces motifs and the opinion-dynamics models of belief updating under repeated interaction. Section 2.3 explains what is meant by an LLM agent and by multi-agent debate, and describes the design space of debate systems. The motifs this thesis detects, and the methodology used to detect them, are defined later in Section 5, as this section focuses on the vocabulary they build on.

2.1 Complex Systems and Self-Organization

A *complex system* is a collection of many interacting components whose collective behaviour cannot be inferred from the properties of the individual components alone (Anderson, 1972, Ladyman et al., 2013). The behaviour is called *emergent*. For instance, a stock-market bubble is produced by millions of traders acting on local information with no coordinating authority, yet the resulting price dynamic is not present in any individual trade (Mitchell, 2009). When such emergent order arises without any central controller, the system is called *self-organizing* (Ashby, 1947, Camazine et al., 2001). Four ingredients are shared of self-organizing systems (Camazine et al., 2001). *Positive feedback* amplifies small changes and drives the system away from uniformity. *Negative feedback* prevents runaway growth and bounds the system within reachable states. *Fluctuations* are the stochastic perturbations that allow the system to explore the state space. *Repeated local interactions* let global structure accumulate over many iterations from purely local rules. The organising principle is fully encoded within these four interactions, not in any single component and also not in any external authority. Thus, the same local rule can lead to different long-run outcomes depending on where the system starts and which fluctuations occur. The interaction rule determines which end states are possible. The starting configuration and the random noise then decide which of them the system actually reaches.

To study self-organizing systems rigorously, one needs a language for how states evolve over time. That language is called *dynamical systems theory* (Strogatz, 2015). A dynamical system consists of a state, which collects everything needed to describe the system at a given moment, and an update rule that turns the current state into the next one. The set of all possible states is the *phase space*, and as time advances the system traces a *trajectory* through it. The classical theory assumes a deterministic rule acting on a continuous state, but three extensions cover the cases that matter here. In a *stochastic* system the rule assigns probabilities to the possible next states instead of fixing one, so the trajectory becomes a Markov chain. In a system with *memory* the next state depends on a window of recent states rather than on the current one alone. In a *discrete-state* system the state takes one of finitely many values, so that the rule becomes a table of transitions. The debate systems studied in this thesis combine all three: the state is discrete, since each agent casts one of finitely many votes; the update is stochastic, since agents sample their

responses; and it carries bounded memory, since each agent conditions on a finite context window of recent rounds.

The central question in dynamical systems theory is where trajectories end up. After an initial phase, most trajectories settle into specific regions of the phase space called *attractors*, towards which nearby trajectories converge and in which they then remain. The classical image is a ball rolling under gravity on a hilly landscape. Wherever the ball starts, friction brings it to rest in one of the valleys. However, not every resting point is an attractor. A ball balanced on a hilltop is also at rest, but it is unstable, because any small push moves it away rather than back. Thus, an attractor is *stable* if small deviations from it decay. The set of starting points from which the system reaches a given attractor is that attractor’s *basin of attraction*. The simplest and most common long-run regime is *fixed-point convergence*, in which the system settles into a single state and stays there. A structurally different regime is the *limit cycle*, a closed loop of states that the system revisits indefinitely without ever settling. A limit cycle is itself attracting as even though a small perturbation pushes the trajectory off the loop, the dynamics pull it back briefly afterwards. In the stochastic, discrete setting of a debate, these two regimes take the form of absorbing states, which are never left once entered, and of sets of states that keep being revisited.

Things get more interesting when the landscape has several attractors instead of only one, as shown in Figure 1. A system is *multistable* when two or more distinct attractors coexist under the same parameters (Feudel, 2008). Figure 1 shows the simplest example, a double-well with two stable states separated by an unstable peak. The boundary between the two basins of attraction is the *separatrix*. Two runs of the same experiment with identical parameters, but starting on opposite sides of the separatrix, end in entirely different long-run states. When many runs of the same system are observed, the share of runs that reach each attractor therefore reflects the relative size of the basins. The double-well is deterministic and continuous, whereas the debate we study is a finite-state stochastic chain. There, coexisting attractors become competing absorbing states, and basin size is no longer a geometric area but a probability defined by the share of repetitions that end in each absorbing state, given how the debate is initialised (Menck et al., 2013).

2.2 Motifs and Opinion Dynamics

The patterns that arise in dynamical systems are not arbitrary. Across biological networks, electronic circuits, social groups, and neural dynamics, the same recurring organisational building blocks appear with consistent functional properties. Complex-systems calls these recurring patterns *motifs*. The concept was introduced for networks by Milo et al. (2002), who defined a *structural motif* as a small subgraph that appears significantly more often in a real network than in a randomly rewired surrogate with the same degree sequence. A classic example is the *feed-forward loop*, a three-node pattern in which A influences both B and C, while B also influences C. In transcriptional networks, it filters out brief signals and delays responses, and similar patterns appear in signalling and neuronal networks (Alon, 2007). A complementary concept is the *dynamical motif* (Driscoll et al., 2024)

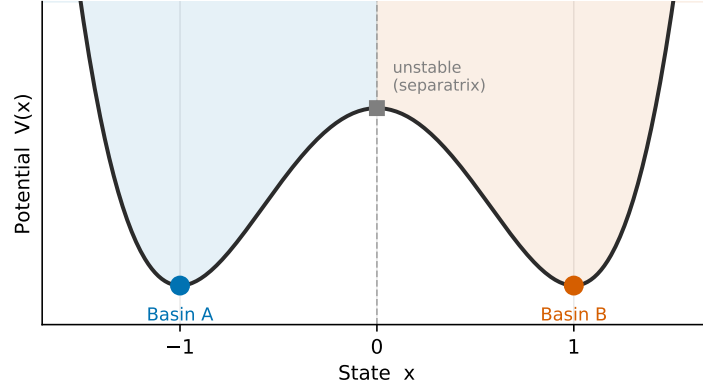


Figure 1: The potential $V(x) = (x^2 - 1)^2$ has two stable attractors at $x = \pm 1$, separated by an unstable fixed point at $x = 0$. The shaded regions are the two basins of attraction. Initial conditions in Basin A converge to the left attractor; those in Basin B converge to the right attractor. Which long-run state the system reaches depends on its initial condition and any fluctuations large enough to cross the separatrix.

which is a recurring pattern in the *trajectory* of a system rather than in its wiring. Where a structural motif is a statement about who interacts with whom, a dynamical motif is a statement about how the system behaves over time. The two are related, since structural motifs are studied precisely because each generates a characteristic dynamical regime, but they are not the same. The same structural motif can produce different dynamics under different parameters, and the same dynamical pattern can arise from different structural configurations.

Positive feedback is a mechanism that links structural and dynamical motifs where small changes are amplified rather than corrected, turning an early advantage into a self-reinforcing lead. Arthur (1989) gave the canonical social example with technology lock-in, where increasing returns keep an early leader ahead even when alternatives are objectively superior. Positive feedback can also operate through network structure. A node positioned to receive and relay information from many others accumulates causal weight, concentrating influence through its position rather than through any informational advantage. Understanding whether such concentration is a property of the topology or of the dynamics themselves requires distinguishing structural connectivity from effective connectivity (Friston, 2011). *Structural connectivity* is the static graph encoding who is allowed to communicate with whom. *Effective connectivity* is how strongly one agent actually affects another during a recorded run. It can be estimated from that run with a model of how influence works, even though the prescribed wiring stays fixed. The two can therefore differ. In a graph where every agent may talk to every other, a few agents can still end up with most of the influence, and which ones do depends on how the run unfolds.

For the specific setting of agents forming and updating beliefs through repeated local interaction, there is a dedicated family of formal models called *opinion dynamics*. The models differ in what they assume about the opinion variable (continuous or discrete),

the update rule (deterministic or stochastic), and who listens to whom (all peers, similar others, or no one). The foundational DeGroot model (DeGroot, 1974) considers N agents, each holding a real-valued opinion, who at every step replace their opinion with a weighted average of the opinions of the agents they trust. The model contains no ground truth and no external information. When the trust network is strongly connected, the process converges to a consensus that is itself a weighted average of the initial opinions. The weight an agent’s initial opinion carries in that consensus is its *social power* (Jia et al., 2015). An agent gains social power by being trusted by other high-power agents, not by being correct. The extreme case, in which almost all social power rests on a single agent while the others carry almost none, is the *autocratic* pole of the model (Jia et al., 2015). It is the opinion-dynamics form of the dominance motif we detect. Convergence to a single consensus requires the trust network to be strongly connected and opinions not to just swap back and forth, and each condition can fail in a way that maps onto one of our four motifs. For instance, if the network contains two or more closed groups that listen only to each other, those groups settle on different consensus values, which is a group-level multistability. On contrast, if opinions keep swapping back and forth and the process never settles but cycles through a repeating sequence of states, a limit cycle emerges.

However, the DeGroot model is too simple for LLM debates in three ways, and each has a standard extension that we use later. Firstly, agents often hold a graded internal belief but must commit to a discrete public choice. The continuous-opinion, discrete-action (CODA) model (Martins, 2008) captures exactly this split and is the template for our agent state. Secondly, the answers in a debate are often discrete labels, which cannot be averaged. Naming-game models describe how a population of such agents ends up agreeing on a single label (Baronchelli, 2018). Different runs can settle on different labels, which is the behaviour our multistability analysis builds on (Section 6.2.2). Thirdly, some agents never change their mind. Such a stubborn agent acts as a fixed anchor that can pull the group toward its position (Friedkin and Johnsen, 1999). Our devil’s advocate probe is exactly this case (paragraph 6.3).

2.3 LLM Agents and Multi-Agent Debate

A *large language model* (LLM) is a neural network trained to predict the next token in a sequence and aligned to follow instructions through supervised fine-tuning and reinforcement learning from human feedback (Ouyang et al., 2022). An *LLM agent* is an LLM placed in a loop with tools, environment feedback, or other models. In this thesis, the loop is formed by other LLMs. Because an LLM samples its output, two agents given identical prompts can produce different responses.

The standard single-model inference-time alternatives to multi-agent protocols are chain-of-thought prompting (Wei et al., 2022), which improves reasoning by encouraging step-by-step intermediate text from one model, and self-consistency (Wang et al., 2023), which samples several reasoning paths from a single model and returns the majority answer. Both can be applied without additional models or communication rounds and serve as the

principal baselines against which multi-agent protocols are compared. A third alternative is *self-correction*, in which a single model critiques and revises its own answer from self-generated feedback (Madaan et al., 2023). However, without any external feedback such intrinsic self-correction often fails to improve reasoning and can even degrade performance (Huang et al., 2024), which is one motivation for bringing in other agents as a source of critique.

Placing several such agents in a shared loop gives an LLM-based *multi-agent system* (Guo et al., 2024). Much of this work gives each agent a distinct role and splits a task among them, for instance by encoding the stages of a software project as standardised operating procedures (Hong et al., 2024), or by providing general frameworks from which such teams can be assembled (Chen, Su, Zuo, Yang, Yuan, Chan, Yu, Lu, Hung, Qian, Qin, Cong, Xie, Liu, Sun and Zhou, 2024). *Multi-agent debate* (MAD) is a special case of this. All agents answer the same question over the same set of possible answers, so their outputs can be compared, contested, and combined into a single group answer. The agents first propose answers independently and then revise them over several rounds after reading each other’s arguments (Du et al., 2024). The term “debate” also names a different proposal in AI safety, in which two models argue before a human judge to make their reasoning checkable (Irving et al., 2018). We follow the collaborative variant, which has no separate judge. The agents themselves both argue and decide.

MAD is motivated by two ideas. The first is error correction. If agents make partly independent mistakes, discussion lets the group catch them. This gain shrinks when the mistakes are correlated, and beyond a certain level of correlation the majority is no better than a single individual actor (Ladha, 1992). The second idea is that a single model struggles to challenge itself. Once it has settled on an answer, reflecting alone rarely produces a new line of thought, a failure that Liang et al. (2024) call *Degeneration-of-Thought*. Peers can supply the challenge that self-reflection lacks. The independence condition matters most for this thesis, because social influence between agents is exactly what makes their votes correlated.

Across its many variants, a debate can be described by a common sequence of steps (Figure 2a). In *initialisation*, each agent answers the question without seeing the others. In *exchange*, each agent receives some subset of its peers’ outputs. In *revision*, it produces an updated argument and vote. *Termination* decides whether another round of exchange and revision follows, and *aggregation* combines the individual votes into a group answer. Every design choice in a debate system fixes how one of these steps is carried out. A recent systematic review of 141 debate studies organises these choices along three dimensions (Motger et al., 2026). These are who takes part (*participants*), how information flows between them (*interaction*), and how the group reaches its answer (*agreement*).

Participants. Agents may be instances of one model or of several (Chen, Saha and Bansal, 2024), and they may carry distinct personas or functional roles (Chan et al., 2024), including a separate judge that moderates the exchange and decides the outcome (Liang et al., 2024). Such variation is the literature’s main means of introducing deliberate diversity into the group, motivated by the independence assumption above: agents that differ are less likely to share the same errors.

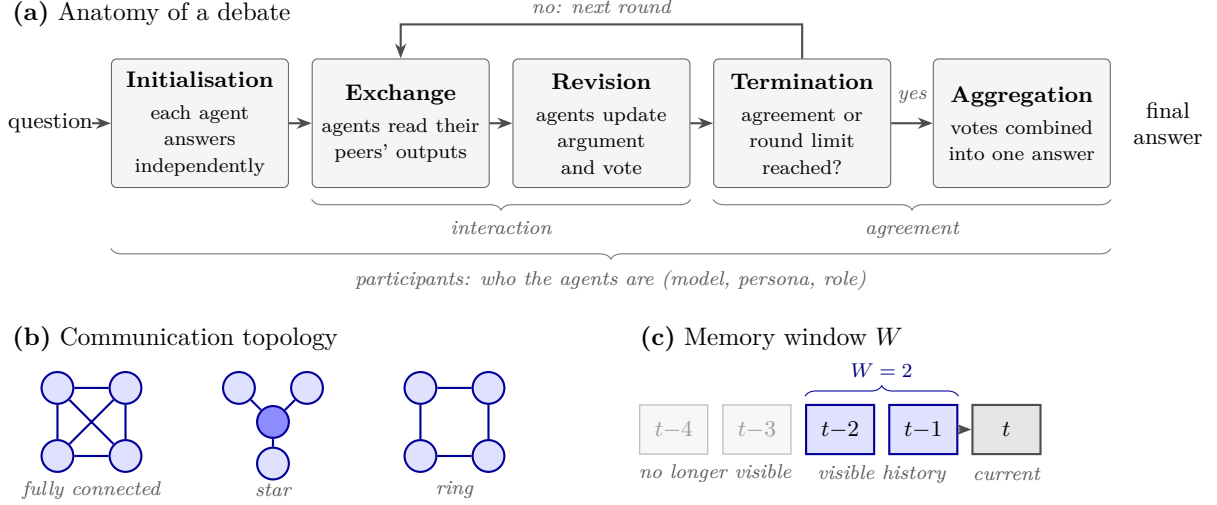


Figure 2: Anatomy and design space of multi-agent debate. **(a)** The generic round structure shared across debate variants. Braces mark the steps governed by the *interaction* and *agreement* dimensions of Motger et al. (2026); the *participant* dimension concerns the agents themselves and applies throughout. **(b)** Three communication topologies for four agents. In the star, all communication passes through the shaded hub. **(c)** A memory window of $W = 2$. At round t , each agent sees only the two preceding rounds. The specific protocol used in this thesis, which instantiates these steps, is shown in Figure 3.

Interaction. The *communication topology* fixes who reads whom (Figure 2b). In a fully-connected group every agent sees every other. In a star, peripheral agents communicate only through a central hub. Sparser arrangements such as a ring restrict each agent to a few neighbours (Li et al., 2024). These topologies are usually fixed in advance, but the communication graph can also be optimised (Zhuge et al., 2024) or adapted while the debate runs (Liu, Zhang, Li, Liu and Yang, 2024). Within a round, agents can speak sequentially in a fixed order or simultaneously, each responding to the same context (Chan et al., 2024). The *memory window* fixes how much interaction history each agent retains from one round to the next (Figure 2c), controlling the influence of earlier rounds on current decisions.

Agreement. The *decision protocol* fixes how individual votes are aggregated into a group answer, for example by majority vote, by a vote weighted by each agent’s stated confidence (Chen, Saha and Bansal, 2024), by a judge (Liang et al., 2024), or by requiring consensus (Kaesberg et al., 2025). Tied to it is the *stopping rule*, which runs the debate for a fixed number of rounds or ends it early once the group agrees or its answers stabilise (Yin et al., 2023, Hu et al., 2025).

The same review finds that the field has converged on a narrow default of static, fully-connected topologies, agents sharing their full messages, short-term memory, and vote-based resolution, adopted by convention rather than through systematic comparison (Motger et al., 2026). The system studied in this thesis (Section 4) stays close to this default, so that its dynamics speak to debate as it is commonly deployed. It departs from it where

measurement requires, by separating private beliefs from public messages, and it varies the two interaction choices shown in Figure 2b and c, topology and memory window. How these design choices behave in practice, and the growing evidence that debate can also amplify errors rather than correct them, is the subject of the related work reviewed next (Section 3).

3 Related Work

This chapter gives an overview of related literature of this thesis. It begins with Section 3.1 and Section 3.2 that survey what multi-agent debate does and how the interaction dynamics inside it have been studied. Section 3.3 continues by weighing the available ways of analysing such systems against one another and Section 3.4 finishes by stating the research gap this thesis addresses.

3.1 Multi-Agent Debate: Design and Critique

The multi-agent debate (MAD) paradigm was introduced by Du et al. (2024), who let several instances of a language model independently propose answers and then critique and revise them over multiple rounds before committing to a collective response. This “society of minds” setup was reported to improve factual accuracy and mathematical reasoning while reducing hallucination, treating the underlying model as a black box. The core intuition was that partially independent agents make partially independent errors which deliberation can correct. Since then it has motivated a broad family of systems, from confidence-weighted round tables across heterogeneous models (Chen, Saha and Bansal, 2024) to evaluation panels that judge text more reliably than a single evaluator (Chan et al., 2024).

Two design axes are common throughout and are the ones we vary too. Communication topology fixes who reads whom. Hereby, sparse structures can match fully-connected debate at substantially lower token cost (Li et al., 2024), and across arbitrary networks the placement of a hub node shifts performance appreciably (Regan et al., 2024). Memory fixes how much interaction history each agent retains. The two were recently crossed in a single setup by Mehdizadeh and Hilbert (2026), who find no main effect of topology on coordination success but a strong interaction with memory. Longer memory slows convergence in decentralised networks and accelerates it in centralised networks, so the same parameter pushes the system in opposite directions depending on the structure of the network. Faster settling in the centralised case means locking into a fragmented plateau rather than reaching agreement. Beyond these two, the decision protocol itself matters, with voting favouring reasoning tasks and consensus favouring knowledge tasks, and adding agents helping where adding rounds can hurt (Kaesberg et al., 2025).

Against this largely optimistic literature, also a more critical literature has emerged. Zhang et al. (2025) find that MAD frequently fails to outperform chain-of-thought (Wei et al., 2022) or self-consistency (Wang et al., 2023) despite far greater compute, and argue that model heterogeneity rather than debate is what reliably helps. Furthermore, Choi et al. (2025) show that majority voting alone accounts for most of the attributed gains, and prove that under their setup debate does not improve expected correctness on its own, since agents’ beliefs change from round to round but on average do not move toward the correct answer. Estornell and Liu (2024) supply the mechanism. When agents are similar in capability or in the responses they produce, debate dynamics become

static and the procedure converges to the majority opinion, so that where the majority reflects a misconception shared through common training data, debate converges to the misconception.

Two 2026 results push this further and are the closest published statements of what we measure. Oh et al. (2026) decompose the failure into a biased static initial belief and homogenised dynamics that amplify the majority view regardless of correctness. Bertalanič and Fortuna (2026) show that unguided homogeneous debate matched or underperformed *isolated self-correction* at several times the token cost, and that a control injecting rationales from unrelated problems performed no worse than genuine peer exchange. They name the resulting loss an *oracle gap*, the distance between the accuracy achievable from the team’s own generation pool and the accuracy the vote actually returns, a quantity we measure directly in Section 6.1.

Almost all of this work, whether it supports or questions debate, judges it by its *final outcome*. They largely ignore the deliberative process behind that outcome, and whether two systems with the same accuracy reach it through different dynamics.

3.2 Interaction Dynamics in LLM Collectives

Another area of work looks inside the debate at how positions evolve and influence one another. Each of our four motifs corresponds to a debate phenomenon that this literature documents to a very different degree, and we take them in turn, before turning to the opening vote distribution, which is not itself a motif but the input on which they all act.

Rising confidence. Prasad and Nguyen (2025) stage repeated adversarial debates and find pervasive miscalibration. Models begin overconfident, grow *more* confident that they are winning as the exchange proceeds even without new supporting evidence, and frequently both claim victory at once, which is a logically impossible. Furthermore, this drift does not depend on correctness. Bertalanič and Fortuna (2026) show that an agent’s confidence before communication barely predicts whether it is right, and that peer exchange even further weakens that link. Confidence that keeps rising regardless of correctness is a typical sign of positive feedback, which motivates a dynamical treatment.

Conformity and influence concentration. Wynn et al. (2025) show that debate can *decrease* accuracy over time even when stronger models outnumber weaker ones, as agents abandon correct answers rather than challenge flawed arguments. Sycophancy (Sharma et al., 2024) and social conformity are among the contributing factors they examine. Cho et al. (2025) find the gap between an agent’s own confidence and its perceived confidence in peers governs how likely it is to conform, and Mehdizadeh and Hilbert (2025b) characterise the dose-response as a sigmoid with a model-specific threshold. Cau et al. (2026) identify an aggregate directional asymmetry they call *agreement drift*, which survives in balanced populations and so cannot be reduced to a majority effect. When influence concentrates in a few agents, a leader-follower structure emerges (Chen et al., 2023). The outcome then depends heavily on how competent the central agent is (Han et al., 2026), a dependency our star topology is designed to expose.

Premature lock-in. Chuang et al. (2025) report that role-playing LLM groups converge far more strongly and quicker than human groups do, indicating a systematic bias toward premature agreement. Purely cooperative framings further sharpen this, since with no dissenting pressure a group can collapse into a mutually reinforcing echo chamber. A structurally sharper version is captured by HiddenBench (Li et al., 2026), which adapts the hidden-profile paradigm from social psychology (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012). When the evidence needed for the correct answer is distributed across agents, collective accuracy collapses far below what a single fully-informed agent achieves, because agents lock onto shared information and fail to pool their private knowledge. We adopt HiddenBench as our second benchmark for exactly this reason.

Oscillation and non-termination. Sustained oscillation is almost absent from this literature. A debate that cycles through the same positions without converging is a limit cycle (Strogatz, 2015), yet debate studies typically fix the number of rounds and count any run that fails to settle as “no consensus”, so a cycle is never distinguished from slow convergence. This gap is itself informative, as we find genuine limit cycles to be rare outside the star topology (Section 6.2.4).

The opening distribution. That round-zero answers govern the outcome is stated in several forms. Hu et al. (2025) make it a formal premise, proving debate improvement over majority vote only under the assumption that at least one initial response is generated under the correct concept. Zhu et al. (2026) report that initial answer diversity correlates significantly with final accuracy and propose a diversity-aware initialisation to raise the chance that a correct hypothesis is present at the start. Empirically, Wu et al. (2025) find that an agent starting inside a correct majority is almost perfectly stable while one facing an incorrect majority recovers in only a minority of cases. In every case the opening distribution is observed as it arises rather than constructed, leaving its effect confounded with task difficulty and agent confidence, which is what Section 6.3 is designed to separate.

The vocabulary for describing these phenomena was introduced in Section 2.2. What remains open is whether it is more than an analogy when applied to LLM populations. Recent work suggests it is. Ashery et al. (2025) show that populations of LLM agents spontaneously reach consensus on an arbitrary shared choice, that strong collective biases can arise even when no individual agent has one, and that a committed minority can overturn the majority position. The last effect mirrors the critical-mass threshold known from human groups (Centola et al., 2018). Okawa (2026) adds a mathematical account, adapting a model from physics to debate. It predicts that a collective norm emerges once conformity passes a threshold, although at the small group sizes typical of debate this threshold is smoothed into a gradual transition.

3.3 Ways of Analysing Multi-Agent Systems

Before stating the gap, it is worth making explicit *why* a complex-systems treatment is the appropriate instrument, by situating it against the alternatives. Each approach answers a different question and has different costs.

Outcome evaluation. The dominant approach scores a system by final task performance (Du et al., 2024, Zhang et al., 2025, Choi et al., 2025). It is objective, comparable across systems, and measures what practitioners care about. But an outcome is only a single number summarising a whole trajectory. Two systems with the same accuracy look identical, even if one corrected its errors and the other slid into a shared mistake. Thus, it only says *whether* a configuration worked, not *why*.

Model internals and transcript reading. Opening the models is the wrong level for our question. Consensus, oscillation, and influence concentration are properties of the *group* trajectory, not of any single forward pass (Miehling et al., 2025), and the internals of API-served models are inaccessible anyway. Reading transcripts is more informative, and we use it informally to examine flagged limit-cycle cases (Section 6.2.4). However, on its own it is subjective, does not scale to tens of thousands of repetitions, and cannot be tested against a null.

Influence attribution. Another approach measures how strongly agents influence one another. This is done either with transfer entropy, which tracks how much one agent’s messages predict another’s (Goodall et al., 2026), or by removing agents and observing how the outcome changes (Lu et al., 2026, Cui et al., 2025). Both give stronger estimates of influence than the adoption-based signature we use, but they require access to the model or repeated re-runs. We return to this trade-off in Section 7.4.

Learned diagnostics from logs. A very recent approach trains a predictor on features taken from interaction logs. It is used either to decide whether a question needs debate at all (Fan et al., 2026) or to decide when a majority vote should be overturned (He et al., 2026). This shows that ordinary logs contain enough signal to support useful decisions. However, such predictors do not explain what is happening. They learn that certain feature patterns precede failure, but do not say which dynamical regime the system is in, and they need labelled outcomes for training.

The dynamical approach. Our approach treats the debate as a dynamical system and measures recurring patterns in its state trajectory (Strogatz, 2015, Driscoll et al., 2024). Its advantages address the gaps above. It operates at the group level, scales to the full dataset, needs only interaction logs and no labelled outcomes, and gives each motif a falsifiable signature that can be tested against a principled null. However, it also has disadvantages. For instance, transferring motif concepts from physical and biological systems to an LLM collective is a modelling choice, not a law. The detection thresholds are partly conventional, and a four-agent group is small for some of these measures. Without intervention, the detectors also identify *signatures* and *associations* rather than certified attractors or causes. We adopt this approach because it is the only one that can name and measure the failure modes documented above while remaining applicable to black-box models. We keep its limitations in mind when interpreting every result (Section 7.4).

3.4 Research Gap

The literature above has described the problem but offers no suitable way to measure it. Outcome-based evaluation, whether it supports or questions debate, only measures where a deliberation ends and ignores how it got there. The dynamics literature does look at the trajectory, but documents one failure mode at a time (overconfidence, herding, premature consensus) and explains each through a shared cause such as sycophancy, not through the motifs they resemble. None of these findings is tested against a null model that shows what the same statistic looks like by chance, so a detection rate alone could simply reflect the small, discrete state space.

More importantly, where complex-systems concepts have been applied to LLM collectives, the tasks were chosen to have no correct answer. Ashery et al. (2025) study arbitrary choices where any shared label is equally good. Mehdizadeh and Hilbert (2026) check that the model itself prefers none of their symbols, so that correctness does not confound the dynamics. Cau et al. (2026) choose a non-empirical proposition for the same reason. This makes those studies clean. If the goal is to understand how a group settles on a shared choice, having no better choice available is exactly right. But it leaves open the question a practitioner faces, which is not whether the group locks in but whether it locks in on the right answer.

This thesis takes up that question. It aims to detect self-organization motifs empirically in *truth-valued* deliberation, following the three research questions of Section 1.2. For *measurement*, we define detectors for self-reinforcement, multistability, dominance, and limit cycles. Each is grounded in the literature and has its own null model. We run each configuration many times, which makes it possible to compute attractor statistics across repetitions. Prior work runs each configuration once (Cau et al., 2026) or analyses run-level means without the distribution of end states across runs (Mehdizadeh and Hilbert, 2026), so it cannot measure multistability even in principle. For *characterisation*, we apply the detectors to tasks with a correct answer, so that every dynamical quantity can be split by correctness, across two different information regimes. For *causation*, we set the composition of the opening votes ourselves instead of observing it, which separates its effect from task difficulty and agent confidence. Throughout, we treat the *absence* of a motif as a finding, not a failed measurement.

4 System and Experimental Setup

The premise of this thesis is that a multi-agent debate is a dynamical system whose trajectory carries measurable structure. Turning that premise into empirical claims requires a system built as a *measurement instrument*, that is not exclusively focusing on a high-performing pipeline. Thus, the protocol must expose a discrete vote and a confidence score per agent and per round, with private belief cleanly separated from public message, so that the interaction log is a record of the underlying dynamics. Furthermore, the two structural axes, communication topology and memory window, must be treated as *controlled perturbation axes* so that we can ask which dynamiscs they cause rather than only observe.

4.1 Notation and System Model

We study a fixed group of N language-model agents that answer the same multiple-choice question over a sequence of discrete rounds repeatedly. All agents are instances of a single frozen model f_θ and only differ in the private context each one has seen. Because the same symbols recur throughout the thesis, we fix the notation here, summarized by Table 1.

Symbol	Meaning
N	number of agents ($N = 4$ throughout)
i, j	agent indices, $i \in \{1, \dots, N\}$
t	round index, $t \in \{0, 1, \dots, T\}$
T	hard cap on the number of rounds
r	repetition index (an independent re-run of a task-config cell)
R	number of repetitions per task-config cell
M	number of answer options a task presents, $M \in \{3, 4\}$
v_i^t	vote of agent i at round t , one of the M options
c_i^t	private confidence of agent i at round t , $c_i^t \in \{1, \dots, 10\}$
$\mathcal{N}(i)$	neighbourhood of agent i under the communication topology
W	memory window (rounds of history each agent may see)
u	early-stopping threshold (consecutive unanimous rounds)
s_t	system macrostate at round t (the vote composition, see below)

Table 1: Core notation used throughout the thesis. Rounds are indexed by t and repetitions by r .

Three units of analysis must be kept apart, since they nest inside one another and are easy to confuse. A *round* is a single synchronous time step t within one debate, at which every agent (re)casts its vote and a debate runs for up to T rounds. A *repetition* (indexed by r) is one complete debate on a given task from round 0 to termination, and because sampling is stochastic, the same task run twice yields different repetitions. A *task-config cell* is a single question paired with one configuration (a topology and a memory window),

and we execute R repetitions of each cell. In short: many rounds make up one repetition, and many repetitions make up one task-config cell.

With the notation established, two structural parts complete the system model. The *interaction structure* is a binary adjacency matrix $\mathbf{G} \in \{0, 1\}^{N \times N}$, where $G_{ij} = 1$ means agent i receives agent j 's messages. Self-loops are excluded ($G_{ii} = 0$), so no agent is fed its own output as a neighbour signal, and the neighbourhood of agent i is $\mathcal{N}(i) = \{j \mid G_{ij} = 1\}$. Any topology can be instantiated by supplying a different matrix without changing the agent or round logic. For our experiments we use two topologies:

- **Fully connected:** $G_{ij} = 1 - \delta_{ij}$, so every agent observes all others each round. This maximises information flow and serves as the baseline. With $N = 4$ agents, each round involves $N(N - 1) = 12$ messages.
- **Star:** one hub agent is connected to all $N - 1$ spokes, and the spokes are connected only to the hub. The hub is drawn uniformly at random per repetition, independently of agent indices, so that any observed hub effect is structural rather than an artefact of indexing. Each round involves $2(N - 1) = 6$ messages, half the fully-connected cost. Star therefore trades per-round token cost for less information flow. As the hub is the sole information bottleneck, agreement takes more rounds to propagate.

The second structural part is the system-level state variable. Individual agents are exchangeable as it only matters how many of them hold each option for the system-level analyses, not which agent holds which. We therefore summarise the group at round t by its *vote composition*, the count of agents on each option,

$$s_t = (n_1^t, n_2^t, \dots, n_M^t), \quad \sum_{k=1}^M n_k^t = N, \quad (1)$$

where n_k^t is the number of agents voting for option k at round t . The number of agents N and the number of options M are independent: $N = 4$ throughout, with $(4, 0, 0, 0)$, $(3, 1, 0, 0)$, $(2, 2, 0, 0)$, $(2, 1, 1, 0)$, $(1, 1, 1, 1)$ as the possible macrostates for $M = 4$ tasks. This macrostate is the object on which the system-level motif detectors of Section 5 operate.

4.2 Debate Protocol

The protocol specifies three things, namely the persistent state each agent carries across rounds, the within-round message passing that produces a new state, and the memory window controlling how much past history each agent may see. We follow the multi-agent debate paradigm of Du et al. (2024), in which agents independently answer and then revise over several rounds, but instrument it so that the interaction log exposes exactly the quantities the motif detectors require.

These requirements translate into a specific per-agent state. At each round t , agent i produces the following four fields.

- v_i^t is the discrete **vote**, one of the task’s M options.
- $c_i^t \in \{1, \dots, 10\}$ is a private **confidence**, never transmitted.
- priv_i^t is private chain-of-thought **reasoning**, never transmitted.
- pub_i^t is a **public message**, broadcast to neighbours.

The separation of the public message from the private reasoning and confidence is intentional as it lets us measure whether agents say what they privately hold. Furthermore, the split of vote and confidence follows the continuous-opinion, discrete-action template of Martins (2008), and the public/private channel separation follows the DEBATE design of Chuang et al. (2025).

Building on this state structure, for rounds $t \geq 1$ each agent also returns an *argument* before committing to a belief, given by

$$\text{argument}_i^t = (\text{defense}_i^t, \text{challenge}_i^t, \text{question}_i^t), \quad (2)$$

consisting of a **defense** (one concrete reason the current vote is correct), a **challenge** (a dispute of a specific peer claim, or a concession), and a **question** (a targeted query to a named peer). The argument is visible to neighbours at the start of the *Update* phase but does not itself change any vote. These fields structure the deliberation but are not themselves used quantitatively, but helped constructing a system of higher debate intensity.

Together, the committed state and the argument define the round structure. Each round consists of two sequential phases. Within each phase all agents act in parallel, but the *Argue* phase runs to completion before the *Update* phase begins. Round 0 is asymmetric, as no communication has yet occurred. Figure 3 illustrates the full message flow for two agents over the first three rounds.

- **Round 0 (initialisation, *Update* phase only).** Each agent receives the question and options and produces an initial belief with no peer input:

$$(v_i^0, c_i^0, \text{priv}_i^0, \text{pub}_i^0) = f_\theta(\text{question}, \text{options}).$$

- **Round $t \geq 1$, *Argue* phase.** Agent i reads its self and peer history windows $\mathcal{H}_i^{\text{self}}(t, W)$ and $\mathcal{H}_i^{\text{peer}}(t, W)$ (defined formally below) and returns an argument. Votes are *not* updated in this phase:

$$\text{argument}_i^t = f_\theta(q, \mathcal{H}_i^{\text{self}}(t, W), \mathcal{H}_i^{\text{peer}}(t, W)).$$

- **Round $t \geq 1$, *Update* phase.** Agent i additionally reads its own and its neighbours' current-round arguments, then updates its full state. If it changes its vote, it is instructed to cite the specific argument that caused the change:

$$(v_i^t, c_i^t, \text{priv}_i^t, \text{pub}_i^t) = f_\theta(q, \mathcal{H}_i^{\text{self}}(t, W), \mathcal{H}_i^{\text{peer}}(t, W), \{\text{argument}_k^t\}_{k \in \{i\} \cup \mathcal{N}(i)}).$$

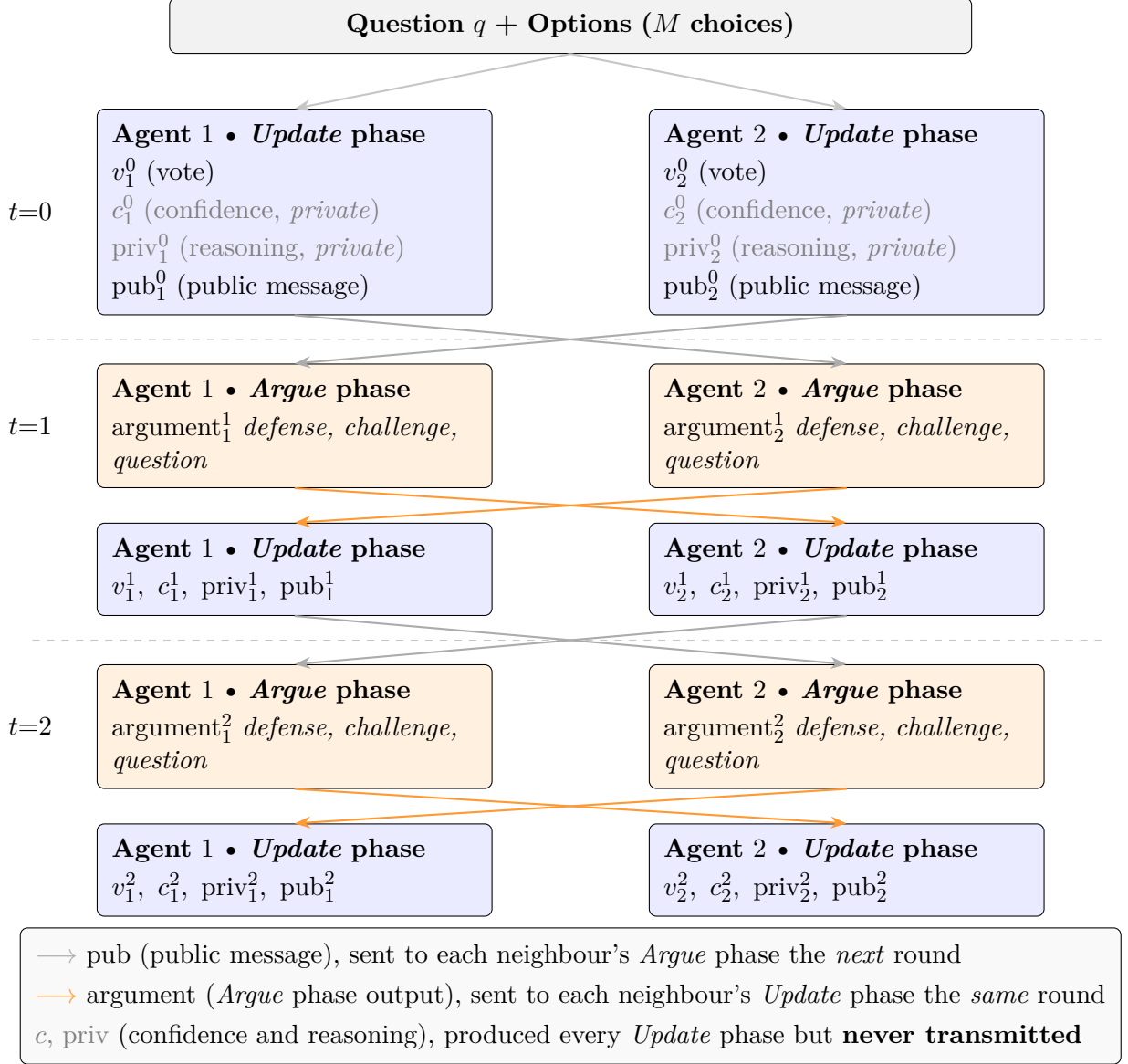


Figure 3: Two-phase synchronous protocol illustrated for two agents over rounds $t=0, 1, 2$. As round 0 has no *Argue* phase, the agents read the question directly. Each *Update* phase commits the four state fields ($v, c, \text{priv}, \text{pub}$), with the confidence c and reasoning priv staying private (grey). A public message reaches neighbours in the *next* round's *Argue* phase, while an *Argue* argument reaches neighbours in the *same* round's *Update* phase, which is what lets one round carry a full question–answer exchange.

The two-phase structure compresses a question–answer exchange into a single round as

a one-phase synchronous protocol could not achieve this without a two-round lag. For the dynamical analysis, only the committed *Update* state is treated as the system state at round t and the arguments mainly function as message memory and debate facilitator. All outputs are returned as structured JSON and no problems with malformed responses were observed. The complete prompt templates are shown in Section A.3.

One aspect of the round structure left unspecified so far is how much history each agent can access. The memory window $W \in \mathbb{Z}_{\geq 1}$ controls how many past rounds each agent sees. At round t , agent i has a *self* window of its own recent states and arguments, and a *peer* window of its neighbours' recent votes, public messages, and arguments, defined as

$$\mathcal{H}_i^{\text{self}}(t, W) = \left\{ \left(v_i^{t-k}, c_i^{t-k}, \text{pub}_i^{t-k}, \text{argument}_i^{t-k} \right) \right\}_{k=1}^{\min(t, W)}, \quad (3)$$

$$\mathcal{H}_i^{\text{peer}}(t, W) = \left\{ \left(v_j^{t-k}, \text{pub}_j^{t-k}, \text{argument}_j^{t-k} \right) \right\}_{j \in \mathcal{N}(i), k=1 \dots \min(t, W)}. \quad (4)$$

Peer entries are presented in randomised order at each call to mitigate position bias, and round-0 entries leave out the argument field, which does not exist yet. A larger memory window W exposes more history but also interacts with documented long-context degradation (Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni and Liang, 2024), which is why we treat W as an explicit perturbation axis rather than fixing it implicitly. $W = 1$ is the near-Markov baseline in which each agent sees only the previous round.

Now that the protocol and memory structure are defined, the remaining question is when to stop. Running every repetition for the full T rounds is wasteful once agents have converged, as pilot transition matrices show a self-loop probability of 97–100 % across tasks for the unanimous states, meaning that once consensus is reached, escape is rare. We therefore stop a repetition as soon as u consecutive rounds are unanimously agreed. The threshold u trades efficiency against reliability. If it is too small, a transient consensus may be cut short, and if it is too large, the savings shrink. Analysing pilot repetitions we find that the harmful case, in which stopping would have prevented a later recovery to the correct answer, stays negligible while coverage remains high, and we set $u = 3$ with a hard cap of $T = 15$. We report the details of this efficiency trade-off in Appendix A.2. Terminating debate at consensus is standard practice in the MAD literature (Yin et al., 2023, Yang et al., 2025, Kaesberg et al., 2025).

When a repetition ends, the system's answer is the *plurality* of the agents' votes at the final round, the option held by the most agents. Because a repetition stops only after $u = 3$ consecutive unanimous rounds, every converged repetition ends with all N agents on one option, so the plurality is unambiguous and equals that option. A tie can therefore only happen in runs that hit the hard cap $T = 15$ without converging, which is about 3.8 % of repetitions. In a tie, one of the tied options is chosen at random. This label is not a real group decision, since it only reflects where the unresolved run happened to end. Section 6.2.4 therefore recommends flagging these repetitions as unresolved instead of trusting the reported majority.

4.3 Datasets

Because this work studies motif emergence rather than task-completion performance, our benchmark selection criteria differ from those usual in the MAD literature. A benchmark suitable for motif detection must provide (i) a **discrete shared decision variable**, so that all agents commit to a position in the same finite set each round, (ii) **verifiable ground truth**, so dynamics can be related to correctness, (iii) **calibrated difficulty**, since saturated tasks remove the disagreement that drives deliberation, and (iv) **sufficient initial disagreement**, without which no motif can be distinguished from trivial consensus.

Applying these criteria rules out most existing benchmarks. Saturated reasoning datasets such as GSM8K and MMLU leave no room for deliberation dynamics, while native multi-agent benchmarks are largely incompatible, being either role-specialised or defined over spatial rather than categorical positions. A full survey of the candidates we considered and the reason each was rejected is given in Section A.1. Two benchmarks satisfy all four criteria and are complementary, so we use them throughout this study.

GPQA-Diamond. GPQA-Diamond (Rein et al., 2024) is a graduate-level science question-answering benchmark of 198 questions in biology, physics, and chemistry, written by domain experts and validated through a “Google-proof” protocol that resists retrieval-based shortcuts. It satisfies all four criteria directly. Four-option multiple choice ($M = 4$) provides the discrete decision variable and expert validation gives unambiguous ground truth. At release, frontier models were well below ceiling (Rein et al., 2024), and while the most capable models have since reached $\sim 85\%$ (Tian et al., 2025), mid-tier models remain well below, leaving improvement potential using debate. GPQA-Diamond also has direct multi-agent precedent (Tian et al., 2025, Kenton et al., 2024, Wang et al., 2025). We adopt it as the **primary** dataset. The main downside is that this benchmark has less cumulative MAD literature compared to MMLU or GSM8K, which is however mostly the case as its a more recent benchmark and not because of a methodological change. A second drawback concerns comparability of the accuracy numbers, as published figures refer to strong models on the full benchmark, whereas we use a mid-tier model on only a screened task subset. Our results are therefore not comparable to published frontier GPQA-Diamond numbers, and are not intended to be, as the screening pass deliberately selects the most dynamically contested tasks, where accuracy is low by design.

HiddenBench. Moreover, we use HiddenBench (Li et al., 2026) as a structurally complementary **secondary** dataset. It formalises the hidden-profile paradigm from social psychology (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012), in which the evidence needed for the correct answer is split across agents, so that no single agent can solve the task alone and success requires information pooling. Its value here is as a control for a specific confound. Motifs seen only on GPQA-Diamond might be artefacts of cooperative reasoning over a reliable individual signal, in which RLHF-induced truth-seeking (Ouyang et al., 2022) amplifies convergence. By contrast, under the hidden-profile condition pre-discussion accuracy is very low by construction (Li et al., 2026), so any post-discussion improvement must come from information transfer rather than from individual model

strength. Motifs that appear on both datasets are therefore robust to this confound, while motifs that appear on only one are themselves informative on a dataset-level.

One further structural detail of HiddenBench matters for the analysis. While we use $N = 4$ agents throughout, the number of answer options is task-dependent, $M \in \{3, 4\}$. The agent count N and the option count M are therefore independent quantities and should not be confused. This is why every option-indexed quantity in Section 5 is defined over the task’s actual M options. Extension to mixed-motive negotiation datasets is left to future work.

4.4 Experimental Design

The system exposes many tunable components. We classify each as **scan** (a primary perturbation axis), **fix** (held constant), or **defer** (out of scope). Table 2 summarises the classification and the literature supporting it, and the detailed per-knob justification is deferred to Section A.2.

Of these components, two are varied as primary perturbation axes, chosen because they are the canonical levers for network-motif analysis:

- **Memory window** $W \in \{1, 2, 5\}$, with $W = 1$ the near-Markov baseline. Memory content is an established driver of debate behaviour (Du et al., 2024, Mehdizadeh and Hilbert, 2026, Zargari Marandi, 2026).
- **Topology** $\in \{\text{FC}, \text{STAR}\}$. The fully-connected topology serves as the symmetric baseline, while the star tests for hub-induced dominance, following evidence that centralised structures concentrate influence (Li et al., 2024, Chen et al., 2023, Regan et al., 2024).

Their product gives $3 \times 2 = 6$ configurations per task. The dataset (GPQA or HiddenBench) is classified as a scanned condition in Table 2 because the information condition varies between benchmarks, but it is not a structural axis in the sense of a manipulation applied within a fixed task, so the two benchmarks are instead analysed separately. The full design therefore spans $6 \times 2 = 12$ configuration–dataset cells. All remaining components are held constant to isolate the effect of the two scan axes:

- **Agents** $N = 4$, the smallest value at which the two topologies are structurally distinct and multi-way splits can occur.
- **Rounds** $T = 15$ (hard cap) with early stopping $u = 3$, where T is an observation budget rather than a structural knob, since trajectories can be cropped post hoc but not extended.
- **Model** `mistral-medium-instruct`, a single mid-tier model to isolate topology and memory effects from capability-asymmetry effects of heterogeneous models (Wynn et al., 2025).

- **Sampling** temperature 0.7 with reasoning disabled, resulting in diverse yet focused outputs. For homogeneous agents stochasticity arises solely from sampling (Zhu et al., 2026), so a positive temperature is necessary for any response diversity.
- **Framing** cooperative, with the full *Argue/Update* structured output as the measurement instrument, and synchronous execution to remove agent-order confounds.

Finally, selecting which tasks to run completes the design. Motifs are per-task properties, and running the full grid at high R on every question is infeasible, since at roughly 25s per repetition, the full GPQA-Diamond set across six configurations would take approximately 17.2 days (413 h) at the final $R = 50$, whereas a two-phase design that uses an initial screening process reduces this to approximately 4.5 days (108 h). It begins with a shallow *screening* pass, which runs every task once at a single baseline configuration (FC, $W = 1$, $R = 3$). Only tasks where debate actually occurs *and* the final answer varies across repetitions are kept, because only for these tasks do the dynamics, rather than the model’s prior, decide the outcome. Tasks are therefore excluded if all agents agree at $t = 0$, if every repetition ends in the same answer, or if there is no real disagreement.

Knob	Verdict	Value	Key references
Memory W	scan	$\{1, 2, 5\}$	(Du et al., 2024, Mehdizadeh and Hilbert, 2026, Zargari Marandi, 2026)
Topology	scan	fc, star	(Li et al., 2024, Chen et al., 2023, Regan et al., 2024)
Agents N	fix	4	(Du et al., 2024, Li et al., 2026)
Rounds T	fix	15 (cap), $u = 3$	(Kaesberg et al., 2025, Yang et al., 2025)
Output structure	fix	full <i>Argue/Update</i>	(Liang et al., 2024, He and Feng, 2025)
Framing	fix	cooperative	(Liang et al., 2024, Du et al., 2024)
Synchrony	fix	parallel-synchronous	(Du et al., 2024, Chan et al., 2024)
Model	fix	mistral-medium-instruct	(Zhang et al., 2025, Wynn et al., 2025)
Temperature	fix	0.7, no reasoning	(Zhu et al., 2026)
Information cond.	scan (dataset)	GPQA ($M=4$), Hidden-Bench ($M \in \{3, 4\}$)	(Li et al., 2026, Stasser and Titus, 1985)
Tool access	defer	none	(Rein et al., 2024)

Table 2: Classification of tunable components. Full justification per knob is given in Section A.2.

Retained tasks are promoted to a deep *characterisation* pass on the full six-configuration grid. This yields the final dataset used throughout Section 6 and consists of **50 tasks**

for each of the two benchmarks, six configurations each, $R = 50$ repetitions, for a total of $50 \times 2 \times 6 \times 50 = 30,000$ repetitions. GPQA-Diamond yields more eligible tasks than needed in the screening pass, so we retain 50 to match the number of eligible HiddenBench tasks, giving a balanced design with equal task counts per benchmark. The 50 are drawn uniformly at random from the eligible GPQA tasks, so that no further selection is made on any outcome-derived quantity. Because $R = 3$ in the screening pass introduces some noise, assignments near the inclusion boundary can be a bit unstable, which we accept as a screening cost rather than a final claim.

This design has two consequences for how the results should be read. First, this affects RQ2 as only tasks selected by the screening pass are run, so task difficulty is roughly constant by construction. It is a *selection criterion* rather than a factor we vary. The “difficulty regime” contrast in RQ2 is therefore made at the benchmark level, between GPQA (a refinement regime, where agents mostly start correct) and HiddenBench (a correction regime, where agents start systematically wrong). Second, this affects the multistability results. Tasks are kept because their final answer varies across repetitions, so the share of tasks with coexisting attractors in Section 6.2.2 only holds for this selection. It should not be read as the frequency of motifs in all tasks.

For more information, code, prompts, and run scripts are all available at <https://github.com/echtermeyer/LLM-Based-MAS>.

5 Detection Methodology

Each of the four motifs defined in this section is a hypothesis about the underlying debate dynamics. If the system self-organizes in the way a complex-systems reading predicts, then confidence should reinforce within a held position, repeated repetitions should settle into a small number of initial-condition-selected attractors, influence should concentrate on particular agents, and in some configurations vote sequences may oscillate persistently without converging. This section turns each hypothesis into a *detector*: a statistic computed from the interaction log together with a principled null model against which it is tested, so that a positive result reflects genuine structure rather than the chance regularities of a small, discrete state space. Together the four detectors form SOMA, the detector suite this thesis contributes.

Two properties are shared by all four detectors and therefore are worth stating up front. Firstly, each detector reads only from ordinary interaction logs, the vote and confidence channels defined in Section 4, which is what later lets us propose the same detectors as diagnostic instruments. And secondly, each detects a *signature*, not a certified dynamical object. Thus, we add a paragraph stating the limitations to each detector which makes explicit what the corresponding detector can and cannot claim.

5.1 Shared Measurement Conventions

Before defining each detector individually, it is worth fixing what they share. Four things recur across every subsection below: the shape those subsections take, the classification of the motifs themselves, the macrostate the system-level detectors read from, and the nesting of the data that constrains every comparison.

The four subsections each make the same five parts, in the same order. *Concept* states the motif in dynamical-systems terms and names the quantity in a debate that plays the role of the state variable. *Measurement* defines the *measurement primitive*, the lowest-level object the detector identifies, and the statistic computed from it. *Detection* specifies the null model against which that statistic is tested, and the rule for flagging the motif. *Worked example* runs the detector by hand on a small case, and *Limitations* states what the resulting statistic cannot claim. Furthermore, so that the four worked examples can be read against one another, they share a single running scenario, one task with $N = 4$ agents and $M = 4$ options labelled A to D of which A is correct, and they write an agent's state as its option with its confidence as a subscript, so that D_8 is a vote for D with a confidence score of 8.

The motifs those subsections detect are preselected rather than exhaustively enumerated and each is motivated by a documented failure mode of LLM-based debate. Table 3 lists them together with their classification and the section in which each detector is defined. Further motifs that fit the same framework but fall outside this study are shown in Appendix A.10.

Motif	Classification	Level	Section
Self-reinforcement	Structural (positive autoregulation)	Agent	Section 5.2
Multistability	Dynamical (coexisting attractors)	System	Section 5.3
Dominance	Structural (emergent hub)	System	Section 5.4
Limit cycle	Dynamical (periodic orbit)	Agent & System	Section 5.5

Table 3: The four motifs classified by type (structural or dynamical, following Section 2.2) and level (whether the detector reads from individual agent trajectories, the collective macrostate, or both).

What the table calls the *level* of a motif decides where its detector reads from. Self-reinforcement, the only agent-level motif, reads directly from per-agent vote and confidence trajectories. The three system-level readings instead operate on the macrostate s_t defined in Section 4.1. It forgets which agent holds which vote but keeps how the votes are split across the options. A 3–1 split therefore differs from a 2–2 split, and a vote for the correct option differs from a vote for a wrong one. However, each of the three motifs takes a different slice of that record. Multistability reads only the locked-in endpoint, dominance reads the emergent influence graph, which is built relationally and therefore needs within-repetition agent identity, and the system-level limit-cycle test reads the full sequence $s_0, \dots, s_T$.

Whichever slice a detector takes, it inherits the same nesting of the data,

$$\text{agent} \subset \text{repetition} \subset \text{task} \subset \text{config} (W \times \text{topology}) \subset \text{dataset}, \quad (5)$$

with $W \in \{1, 2, 5\}$, $\text{topology} \in \{\text{fc}, \text{star}\}$, $N = 4$, and $\text{dataset} \in \{\text{GPQA}, \text{HiddenBench}\}$, and that nesting constrains what may be compared with what. Because agents interact within a repetition, their trajectories are not independent, so the repetition is the independent unit of replication and agent-level trajectories are never pooled as if they were separate observations. For the same reason GPQA and HiddenBench are always analysed separately and never pooled, and memory-window comparisons are made within a fixed topology and dataset, since cells from different configs are not exchangeable.

All four detectors share the limitation that they are descriptive and not causal. They find a *signature* in the trajectories or influence graph, not a proven attractor, cycle, or causal hub. Telling the two apart would need perturbation experiments, which we leave to future work. Because the debates are cooperative and forced to converge, we also expect several detectors to find little or nothing. This is why each detection paragraph below compares its statistic against a named baseline rather than against zero.

5.2 Self-Reinforcement

Concept. In complex systems, self-reinforcement is *positive feedback* reinforcing itself. The current level of a state variable raises the drive toward more of the same, so small changes are amplified rather than damped (Arthur, 1989). At the motif level this is

positive autoregulation, a node that augments its own production (Milo et al., 2002, Alon, 2007). For us the state variable is an agent’s expressed *confidence*. Self-reinforcement is then the tendency of an agent to push its own confidence upward while it holds a position, driven by accumulating commitment rather than by new incoming information. The pattern is already documented as in multi-turn LLM debate, confidence tends to rise across rounds even where no new information justifies the rise (Prasad and Nguyen, 2025).

Rising confidence alone is not enough to show self-reinforcement, since rational updating and peer conformity also raise confidence. We therefore measure two further signatures. *Directional drift* is confidence tending to rise rather than fall while an agent holds its position. *Strength scaling* is this drift growing the more an agent is re-exposed to its own earlier commitments, which the memory window W controls directly. A third signature, *endogeneity*, would show that the rise comes from the agent’s own commitment rather than from new information, but trajectory data cannot separate it from peer effects.

Measurement. The primitive is the *vote-stable segment* which is the longest block of consecutive rounds, from t_s to t_e , in which agent i keeps the same vote, so that $v_i^t = v_i^{t_s}$ for all $t \in \{t_s, \dots, t_e\}$. With $L = t_e - t_s + 1$ as its length in rounds, we keep a segment only if $L \geq 3$. Three rounds give two steps for the least-squares fit to average over. An agent that flips can contribute several segments, and one that locks in early contributes a single long one. Agents are exchangeable, so we pool segments within a config and never match them across repetitions. Within each kept segment we fit the confidence on the round index by ordinary least squares,

$$c_i^t = \beta_0 + \beta_{\text{seg}} t + \varepsilon_t, \quad t \in \{t_s, \dots, t_e\}, \quad (6)$$

and keep the slope β_{seg} , in confidence points per round. Its sign enters the prevalence test below, and its value enters the mean slope $\bar{\beta}$ used in the W comparison.

Detection. To detect self-reinforcement we ask whether segment slopes point upward more often than chance would allow. We read them in two ways, both within a single config. The first is a presence test for directional drift. Let n_{seg} be the number of kept segments in a config-task pair. The share of upward segments is

$$\hat{p}_{\text{SR}} = \frac{|\{\text{seg} : \beta_{\text{seg}} > 0\}|}{n_{\text{seg}}}, \quad (7)$$

so flat segments ($\beta_{\text{seg}} = 0$) count in the denominator but never in the numerator. Under a no-feedback “wobble” null, an upward and a downward segment are equally likely. With a fraction f of flat segments this gives $P(\beta_{\text{seg}} > 0) = \frac{1}{2}(1 - f) \leq 0.5$. We therefore take $\hat{p}_{\text{SR}} > 0.5$ as an indicator of self-reinforcement. Flat segments and the confidence ceiling both make a positive result harder to get, because a segment already at the top of the scale cannot drift further up. The second reading uses the memory window to tell the mechanisms apart. Let $\bar{\beta}$ be the mean segment slope in a config. An *endogenous* reading predicts that re-reading the own earlier messages over a longer window strengthens the held position,

$$\bar{\beta}(W=5) > \bar{\beta}(W=2) > \bar{\beta}(W=1), \quad (8)$$

whereas a *peer-conformity* reading expects a longer window to mainly add peer content, so the ordering may be missing or reversed.

We report both readings per config-task and then summarise them across tasks, for example “above 0.5 in k of 50 tasks”. This shows whether an effect is broad or driven by a single task.

Worked example. In the running scenario, suppose agent 0 holds option A for five rounds with confidences $c = (5, 6, 6, 8, 9)$. This is one vote-stable segment of length $L = 5$. Regressing confidence on the round index gives $\beta_{\text{seg}} = +1.0$ confidence points per round, so the segment is upward and counts toward $\hat{p}_{\text{SR}}$. Had the agent instead drifted $c = (7, 7, 6, 6, 5)$, the slope would be -0.5 and the segment would count against it. A segment that never moved, $c = (6, 6, 6, 6, 6)$, has $\beta_{\text{seg}} = 0$. It stays in the denominator but is not counted as upward, which is what makes the 0.5 reference conservative. Under the strength-scaling reading, the average of such slopes should grow as the window widens from $W=1$ to $W=5$.

Limitations. Beyond the shared problem that the detectors are descriptive, two points limit the detection. First, *endogeneity is not isolated* as an upward segment fits equally well with peer copying and with a peer adding genuinely new information in a later round, and rising confidence alone cannot tell these apart. The W ordering narrows this ambiguity but does not settle it, since a longer window also re-exposes more peer content. Second, *convergence is a confound* as on easier tasks, fast agreement can itself raise confidence. This is why we report prevalence per task and not as a single pooled figure.

5.3 Multistability

Concept. In dynamical systems, *multistability* is the coexistence of two or more stable attractors for one fixed parameter set. Which attractor the system settles into depends on the basin of attraction that contains the initial condition (Feudel, 2008, Pisarchik and Feudel, 2014). At the motif level this is the dynamical signature of positive feedback with a symmetry. The same interaction rules allow several self-consistent outcomes (Milo et al., 2002, Alon, 2007). For us the state variable is the macrostate s_t , and multistability is read off the *locked-in endpoint*. For a fixed task, the debate converges to different consensus answers on different repetitions, and the initial vote configuration, rather than the task’s correct option, decides which one is reached. This is the structural reading of unreliable correction. On a task where the model carries a systematic bias, a wrong-consensus and a right-consensus attractor can coexist. The group’s fate is then a competition between basins decided at round 0, which self-reinforcement amplifies and locks in. Seeing more than one final answer is necessary but not sufficient, as a task may simply be noisy. Genuine multistability additionally requires that the initial condition *predicts* the endpoint (Pisarchik and Feudel, 2014).

Measurement. The primitive and independent unit is the repetition, described by its initial vote composition $s_0 = (n_1^0, \dots, n_M^0)$ and its final macrostate. A repetition is converged

if its final macrostate is unanimous, whether that was reached by early stopping or at the round cap T . We analyse only converged repetitions, because a stable non-unanimous endpoint means the group failed to agree and is not a genuine attractor. Repetitions that end without unanimity are excluded. The endpoint attractor $A^{(r)}$ of a converged repetition is the winning option, so each task has at most M distinct attractors. We score each task-config cell separately, with n_c the number of converged repetitions in that cell. The cell’s *dominant attractor* is its most frequent endpoint option. Each repetition’s start is summarised by

$$g(s_0) \in \{0, 1, \dots, N\}, \quad (9)$$

the number of agents whose round-0 vote is on the dominant option. From these we compute the empirical basin distribution over endpoint attractors, $\hat{P}(A = k) = \frac{1}{n_c} \sum_r \mathbf{1}[A^{(r)} = k]$. This is a basin-stability estimate of relative basin volumes under the model’s own initial distribution (Menck et al., 2013). The effective number of coexisting attractors is the inverse Simpson (Hill) index (Hill, 1973),

$$N_{\text{eff}} = \frac{1}{\sum_k \hat{P}(A = k)^2} \in [1, M]. \quad (10)$$

A value of $N_{\text{eff}} \approx 1$ means a single attractor, ≈ 2 means two coexisting attractors, and so on. Attractors are unanimous options, so N_{eff} is bounded by M . Since M varies across tasks, we report N_{eff} raw and compare it only within a fixed M . GPQA always has $M = 4$, while HiddenBench mixes $M = 3$ and $M = 4$, so we report its N_{eff} summaries separately by option count.

Detection. Coexistence alone does not establish multistability. The initial condition must also *select* the attractor (Feudel, 2008, Pisarchik and Feudel, 2014), and this association carries the inferential claim. We measure it between $g(s_0)$ and the endpoint A with Cramér’s V (Cramér, 1946) on their contingency table,

$$V = \sqrt{\frac{\chi^2/n_c}{\min(R_{\text{tab}} - 1, C_{\text{tab}} - 1)}} \in [0, 1], \quad (11)$$

where χ^2 is the Pearson statistic and $R_{\text{tab}}, C_{\text{tab}}$ are the table’s numbers of rows (distinct $g(s_0)$, at most $N + 1$) and columns (distinct endpoints, at most M). A value of $V \approx 0$ means the start is irrelevant, and V near 1 means the start fully determines the endpoint. The null shuffles the pairing of $g(s_0)$ and A across the n_c repetitions. This keeps both marginals but destroys the link between start and endpoint (Theiler et al., 1992). With $B = 1000$ surrogates, we flag basin selection when the add-one one-sided p -value is below 0.05.

The two readings combine into a single cell label,

$$\text{label}(c) = \begin{cases} \text{monostable} & N_{\text{eff}} < 1.5, \\ \text{multistable} & N_{\text{eff}} \geq 1.5 \wedge p < 0.05, \\ \text{stochastic (not multistable)} & N_{\text{eff}} \geq 1.5 \wedge p \geq 0.05, \end{cases} \quad (12)$$

and we read the degree of multistability from N_{eff} , capped at M . The cutoff 1.5 is a reporting threshold and not a test, so the inferential claim rests on p . The third case is the essential guard, because coexistence without basin selection is scatter and not multistability. Per-cell power is limited, so we rely on the distribution of N_{eff} and V across tasks rather than on any single cell.

Worked example. In the running scenario, take a cell with $n_c = 30$ converged repetitions whose dominant attractor is option A . The coexistence reading depends only on the endpoint distribution, and the label depends on whether the start predicts it.

- **Monostable.** Endpoints $28 \times A$, $2 \times B$ give $\hat{P} = (0.933, 0.067)$ and $N_{\text{eff}} = 1/(0.933^2 + 0.067^2) \approx 1.14 < 1.5$, so there is one valley.
- **Multistable.** Endpoints $15 \times A$, $15 \times B$ give $N_{\text{eff}} = 2.0$. If repetitions starting with more agents on A (high $g(s_0)$) end on A and those starting with few end on B , then $V \approx 1$ and $p < 0.05$, so the start selects between two basins.
- **Tristable.** Endpoints $10 \times A$, $10 \times B$, $10 \times C$ with the start predicting each give $N_{\text{eff}} \approx 3.0$ and $p < 0.05$, the maximal degree for $M = 3$.
- **Stochastic.** Endpoints $15 \times A$, $15 \times B$ again give $N_{\text{eff}} = 2.0$. If $g(s_0)$ is unrelated to the endpoint, then $V \approx 0$ and $p \gg 0.05$. This is coexistence without basin selection and is correctly *not* labelled multistable.

Limitations. First, classical basin stability draws initial conditions uniformly (Menck et al., 2013), whereas our starts come from the model’s own round-0 distribution. $\hat{P}(A)$ therefore estimates the *effective* basin volume under that distribution and not the geometric volume. Second, $g(s_0)$ counts agents already on the eventual dominant option, so a high V fits both genuine basin selection and ordinary persistence of the initial majority. The test detects dependence of the endpoint on the start but cannot separate these two mechanisms. We therefore treat the association as necessary but not decisive.

5.4 Dominance

Concept. Dominance is the concentration of directed influence onto a single node, an *emergent hub*. This is a structural motif (Milo et al., 2002, Alon, 2007) read at the system level, since it is a property of the whole influence distribution. Its formal home is opinion dynamics. In the DeGroot model (DeGroot, 1974) and its social-power extension, an agent’s influence equals its network centrality. The group is *autocratic* when influence is maximally uneven and *democratic* when it is uniform (Jia et al., 2015). Dominance is the autocratic side. For us the state variable is the distribution of influence across the four agents, and dominance is the tendency of this distribution to concentrate on one agent rather than spread evenly. When it does, a single overconfident agent can pull the rest of the group toward its answer (Chen et al., 2023, Cho et al., 2025), which lowers collective accuracy even when the majority would otherwise be correct (Wynn

et al., 2025). Concentration can also come from the wiring alone, so we measure influence for each topology separately. The result is a concentration signature and not proof of causation, since an adoption edge cannot distinguish genuine influence from coincidental agreement.

Measurement. The primitive and independent unit is the per-repetition influence graph, which requires agent identity within a repetition. Dominance is defined over agents and their conversions, so it does not depend on the option count M . A *conversion* occurs when agent j adopts, at round $t+1$, a vote that a disagreeing neighbour i held at round t . The directed edge $i \rightarrow j$ collects the confidence-weighted conversions of j onto i ’s vote. Credit is split among the neighbours that already held the target vote,

$$w_{ij} = \sum_t \frac{1}{m_j^t} \mathbf{1}[v_j^{t+1} = v_i^t \neq v_j^t] c_j^t, \quad m_j^t = |\{k \in \mathcal{N}(j) : v_k^t = v_j^{t+1}\}|, \quad (13)$$

where c_j^t is the converting agent’s confidence, $\mathcal{N}(j)$ is its neighbourhood in the topology, and the factor $1/m_j^t$ shares credit equally among all eligible sources. Per-agent influence is the weighted out-strength $\sigma_i = \sum_j w_{ij}$. We use out-strength rather than eigenvector centrality (Bonacich, 1972), which is undefined or degenerate on non-strongly-connected and acyclic graphs (Newman, 2010) and concentrates on a few nodes (Martin et al., 2014). Our graphs are tiny and sparse, which makes out-strength the only node score that is always defined.

The statistic is the concentration of the influence-share vector $p_i = \sigma_i / \sum_k \sigma_k$, measured by the normalised Herfindahl index (Hirschman, 1964),

$$D = \frac{\sum_i p_i^2 - \frac{1}{N}}{1 - \frac{1}{N}} \in [0, 1]. \quad (14)$$

A value of $D = 0$ is democratic, with equal influence for all agents, and $D = 1$ is a single pure hub. D is undefined when $\sum_k \sigma_k = 0$, meaning no conversion occurs at all, and we exclude such repetitions.

Detection. A high D in the star topology says little on its own. The centre is the only agent a leaf can be converted by, so the wiring alone forces concentration. We therefore key the null to the topology. It asks how concentrated D would be if influence were assigned at random on the same wiring and the same conversion events. We use a constrained shuffle in the spirit of Theiler’s “Algorithm 0” (Theiler et al., 1992). It keeps the converting agent, the round, the weight, and the number of eligible sources fixed, and redraws only the source *identity*, uniformly among $\mathcal{N}(j)$. In the fully-connected topology every agent has three neighbours, so the shuffle can reassign any conversion to any of them and the null is symmetric. Any concentration above it is therefore emergent. In the star topology, however, a leaf has exactly one neighbour, so a leaf conversion has only one possible source and the shuffle leaves it unchanged. When a repetition consists of such conversions, every surrogate reproduces the observed graph and the test can never flag. The star null is thus degenerate, and we identify emergent concentration only in the fully-connected topology. We still report raw D for both topologies, but in the star it describes the wiring and not the behaviour.

A repetition is concentrated when its observed D falls in the upper tail of its own surrogate distribution. With $B = 1000$ surrogates the add-one one-sided p -value is

$$p = \frac{1 + \#\{b : D^{(b)} \geq D\}}{1 + B}, \quad (15)$$

and we flag the repetition when $p < 0.05$. For the fully-connected topology we also report the standardised excess $z_r = (D_r - \mu_{\text{null}})/\sigma_{\text{null}}$ and compare the flag rate with the 5 % that the threshold produces by chance. We summarise each per task and then count across tasks. As for self-reinforcement, an entrenching hub predicts that concentration grows as the memory window widens at fixed topology. We also report whether the hub’s vote matches the final consensus and how often the hub capitulates.

Worked example. In the running scenario with the fully-connected topology, agent 0 opens at D_8 while agents 1, 2, and 3 open at A_6 , B_5 , and C_5 respectively. The vote sequence is

t	agent ₀	agent ₁	agent ₂	agent ₃
0	D_8	A_6	B_5	C_5
1	D_9	D_6	B_5	C_5
2	D_9	D_7	D_5	C_5
3	D_9	D_7	D_6	D_5

Three conversions occur. At round 1, agent 1 flips to D . Only agent 0 held D at round 0, so the full weight of agent 1’s confidence goes to edge $0 \rightarrow 1$, giving $w_{01} = 6$. At round 2, agent 2 flips to D . Both agents 0 and 1 held D at round 1, so credit splits equally, giving $w_{02} = w_{12} = 2.5$. At round 3, agent 3 flips to D . All three of agents 0, 1, and 2 held D at round 2, so credit splits three ways, giving $w_{03} = w_{13} = w_{23} = \frac{5}{3}$. The out-strengths are therefore $\sigma = (\frac{61}{6}, \frac{25}{6}, \frac{5}{3}, 0)$, the influence shares are $p \approx (0.635, 0.260, 0.104, 0)$, and $D = (0.482 - 0.25)/0.75 \approx 0.31$. Agent 0 accounts for 64 % of influence, yet D is well below 1. The credit-splitting rule makes each early adopter a co-source of all later conversions, so staggered convergence never produces a pure hub. $D = 1$ would require every conversion to trace back to a single agent, which is only possible when every other agent flips simultaneously.

Limitations. First, with four agents the four shares and a discrete null limit how small p can get. Per-repetition power is therefore low and the signal lives in the aggregate. Second, a repetition with a single conversion gives $D = 1$ trivially. We handle this through aggregation and the topology null and do not discard such repetitions. Third, as noted above, the star null is degenerate, so the test for emergent excess is identified only in the fully-connected topology. Any claim about emergent concentration in the star rests on raw D and its W ordering and not on the null comparison.

5.5 Limit Cycles

Concept. In dynamical systems, a limit cycle is an isolated closed orbit, a self-sustained periodic oscillation that needs no external forcing (Strogatz, 2015). At the motif level,

sustained oscillation is the dynamical signature of feedback loops, the complement of the positive autoregulation behind self-reinforcement (Milo et al., 2002, Alon, 2007). This is the only one of our motifs that lives at both agent and system level, so it has two state variables. At the agent level the state variable is the agent’s vote $v_i^t \in \{1, \dots, M\}$, and a limit cycle is an agent that keeps cycling through votes (for example $A, B, A, B, \dots$) without ever settling. At the system level it is the macrostate s_t , and a limit cycle is a collective configuration that cycles without settling. Neither implies the other, so we detect and report both. This matters because convergence to consensus is one possible fate of the system, while sustained oscillation is a different one, a periodic disagreement that never resolves. Detecting it tests whether cooperative debate can become trapped in a cycle and not at a point. With only a few symbols, states repeat by chance, and a transient fluctuation that later converges is not a cycle. The definition therefore requires two things. The oscillation must be *self-sustained*, meaning the sequence does not converge, and its periodic structure must *beat a chance baseline*. We measure both. The isolation and stability of a true limit cycle cannot be certified from one observed trajectory, and we do not claim them.

Measurement. The primitive depends on the level. At the agent level it is the per-agent vote sequence within a repetition, $v = (v_0, v_1, \dots, v_T)$, using the committed *Update* vote at each round. At the system level it is the per-repetition composition sequence $s = (s_0, s_1, \dots, s_T)$, so the repetition is both the measurement primitive and the independent unit. Everything that follows applies unchanged to either level, so we write $x = (x_0, \dots, x_{L-1})$ for whichever symbolic sequence is being scored.

A sequence that converges is not a limit cycle, so we exclude converged sequences before computing anything. A sequence counts as converged when it ends in a constant run of at least $u = 3$ symbols, the settling criterion already used in Section 5.3. We judge this on the sequence itself and not on the debate, for two reasons. An agent can settle while the debate around it continues. A debate that deadlocks at a 3–1 or 2–2 split holds that composition until the cap T , which is a fixed point even though no consensus was reached. Every remaining sequence is a limit-cycle candidate, provided it passes a guard $L \geq 4$ that leaves room for one repeat of the shortest possible cycle, which has period 2. We drop converged sequences whole instead of trimming them, since a trailing constant run cannot be separated into cycle-internal repeats and genuine convergence.

We quantify periodicity in the remaining candidates with categorical recurrence quantification analysis (Coco and Dale, 2014). For sequence positions a, b the recurrence matrix records where the sequence revisits a symbol,

$$R(a, b) = \mathbb{I}[x_a = x_b], \quad a \neq b, \quad (16)$$

and excludes the main diagonal, which is trivially all ones (Eckmann et al., 1987). The pattern of ones tells the cases apart. A periodic orbit produces diagonal lines parallel to the main diagonal, spaced equally by the period. A fixed point produces a solid block, and scattered points mean no periodicity at all (Marwan et al., 2007). Two quantities summarise this. *Determinism* is the fraction of recurrence points on diagonal lines of

length $l \geq l_{\min} = 2$ (Zbilut and Webber Jr., 1992, Webber Jr. and Zbilut, 1994),

$$DET = \frac{\sum_{l \geq l_{\min}} l P(l)}{\sum_{l \geq 1} l P(l)}, \quad (17)$$

where $P(l)$ is the histogram of diagonal-line lengths. The *dominant period* is the lag at which recurrence is densest,

$$RR(\tau) = \frac{1}{L - \tau} \sum_{a=0}^{L-1-\tau} R(a, a + \tau), \quad \hat{P} = \arg \max_{\tau \geq 1} RR(\tau). \quad (18)$$

We need both, because DET alone cannot separate persistence from periodicity. A solid block and equally spaced diagonals raise it equally, so a single stray symbol in an otherwise constant sequence can reach a high DET without oscillating at all. We therefore require $\hat{P} \geq 2$ and cap it at $\hat{P} \leq \lfloor L/2 \rfloor$, since a longer period cannot recur twice within the window.

Detection. The null uses random-shuffle surrogates, which are random permutations of the sequence’s own symbols (Theiler et al., 1992). Shuffling destroys temporal order but keeps the symbol frequencies, so the chance-match rate implied by the alphabet is built into the null. For $B = 1000$ surrogates we compute $DET^{(b)}$ and the add-one one-sided p -value (Phipson and Smyth, 2010)

$$p = \frac{1 + \#\{b : DET^{(b)} \geq DET\}}{1 + B}, \quad (19)$$

and flag a limit cycle when

$$p < 0.05 \quad \text{and} \quad 2 \leq \hat{P} \leq \lfloor L/2 \rfloor. \quad (20)$$

We report a binary flag per candidate, the prevalence per config-task as the flagged share of eligible candidates, and summaries across tasks per topology and W .

The two levels share this machinery and differ only in power. The macrostate space has $\binom{N+M-1}{M-1}$ compositions, 15 for $M = 3$ and 35 for $M = 4$, against only M raw vote symbols. Chance recurrence is therefore lower and DET cleaner at the system level. On the other hand, there is one sequence per repetition and not one per agent, so there are far fewer candidates.

Worked example. In the running scenario, suppose agent 2 fails to settle and cycles $v = (A, B, A, B, A, B)$ across $L = 6$ rounds. It never ends in a constant run, so it is a candidate, and $L \geq 4$. The recurrence matrix (main diagonal excluded) is

$$R = \begin{pmatrix} \cdot & 0 & 1 & 0 & 1 & 0 \\ 0 & \cdot & 0 & 1 & 0 & 1 \\ 1 & 0 & \cdot & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdot & 0 & 1 \\ 1 & 0 & 1 & 0 & \cdot & 0 \\ 0 & 1 & 0 & 1 & 0 & \cdot \end{pmatrix}.$$

The ones lie on diagonals spaced by 2, giving $RR(1) = 0$, $RR(2) = 1$ and $RR(3) = 0$, so $\hat{P} = 2$, which satisfies the guard $2 \leq \hat{P} \leq \lfloor 6/2 \rfloor = 3$. The offset-2 diagonal carries a line of length 4 and the offset-4 diagonal a line of length 2, so $DET = (4 + 2)/(4 + 2) = 1.0$. The null shows why length matters. The multiset $\{3A, 3B\}$ has $\binom{6}{3} = 20$ arrangements, and only the two perfectly alternating ones reach $DET = 1$, so $p = 2/20 = 0.10$. At $L = 6$ even a perfect period-2 cycle cannot clear 0.05. The same pattern over a full-length candidate ($L \approx 16$) is flagged decisively, so only long, non-converged sequences are informative. Two exclusions make this concrete. The sequence (A, B, A, B, B, B, B) ends in a constant run and is dropped as a fixed point. The sequence (A, B, B, B, B, B, B) reaches a high DET but has $\hat{P} = 1$, so the period guard rejects it as persistence and not oscillation. The same scenario shows why we report the two levels separately. Suppose agents 0 and 1 swap $A \leftrightarrow B$ each round while agents 2 and 3 hold their votes. Every s_t is then identical. The agent-level test flags two period-2 cycles, while the system-level test sees a fixed point and excludes the repetition entirely. Conversely, a rotating majority can cycle s_t with no single agent being periodic.

Limitations. First, the alphabet is small at the agent level, which inflates the chance recurrence rate, and at both levels the sequence length caps the detectable period at $\lfloor L/2 \rfloor$. Second, the shuffle null accounts for chance recurrence, but the period guard makes the test conservative, so a small nonzero flag rate remains under a chance baseline. We compare observed prevalence against this floor and not against zero.

6 Results

This section reports results from 30,000 debates across GPQA and HiddenBench. Because debate plays a fundamentally different role on each benchmark, every result is reported per benchmark rather than pooled. Section 6.1 establishes the accuracy and efficiency baseline for the system itself. Section 6.2 measures the four self-organisation motifs against their nulls and links each to accuracy and efficiency, which addresses RQ1 and RQ2. Section 6.3 then tests causally whether the opening vote composition determines the outcome, which addresses RQ3.

6.1 Accuracy and Efficiency

Before examining the self-organisation patterns, we report what the system achieves in aggregate and which design choices shape that outcome. Accuracy is the fraction of repetitions in which the final plurality vote matches the ground truth, and efficiency is the mean number of rounds to consensus. Our reference point for debate gain is the *majority-vote baseline*, which is the plurality of the four agents’ round-0 answers before any exchange occurs.

Dataset	Per-agent	Majority vote ($r=0$)	After debate	Oracle ($r=0$)
GPQA	38.2 %	40.5 %	42.2 %	70.6 %
HiddenBench	14.9 %	7.6 %	26.1 %	46.3 %

Table 4: Accuracy decomposition at round 0 and after debate, pooled over all configurations with $n = 15,000$ per dataset. Per-agent is the mean fraction of correct individual answers at round 0. Majority vote is the accuracy of the round-0 plurality vote and serves as the reference point for all debate-gain calculations. Oracle is the fraction of repetitions in which at least one agent is correct at round 0.

Table 4 shows that the two benchmarks operate in qualitatively different regimes. On GPQA the round-0 majority vote is already 40.5 %, and debate adds a modest +1.7 pp to reach 42.2 %. The oracle bound of 70.6 % shows that the correct answer is present among the agents in roughly two-thirds of repetitions. The resulting oracle gap (Bertalaníč and Fortuna, 2026) is therefore not a knowledge deficit but the group’s failure to identify and amplify the correct voice. HiddenBench is different. Here a shared model bias makes majority voting actively harmful and drags the round-0 plurality well below chance. Debate partly corrects this and reaches 26.1 %, a gain of +18.5 pp. Repetitions that converge early lock in the shared wrong answer almost universally, because early consensus forecloses the corrective exchange that the benchmark design requires, as Figure 12 in Appendix A.4 shows. In summary, debate refines a mostly-correct start on GPQA. On HiddenBench it partially corrects a systematically wrong one.

With the outcome picture in place, the remaining question is what design choices such as memory window and topology contribute. Figure 4 plots accuracy and rounds across all configurations.

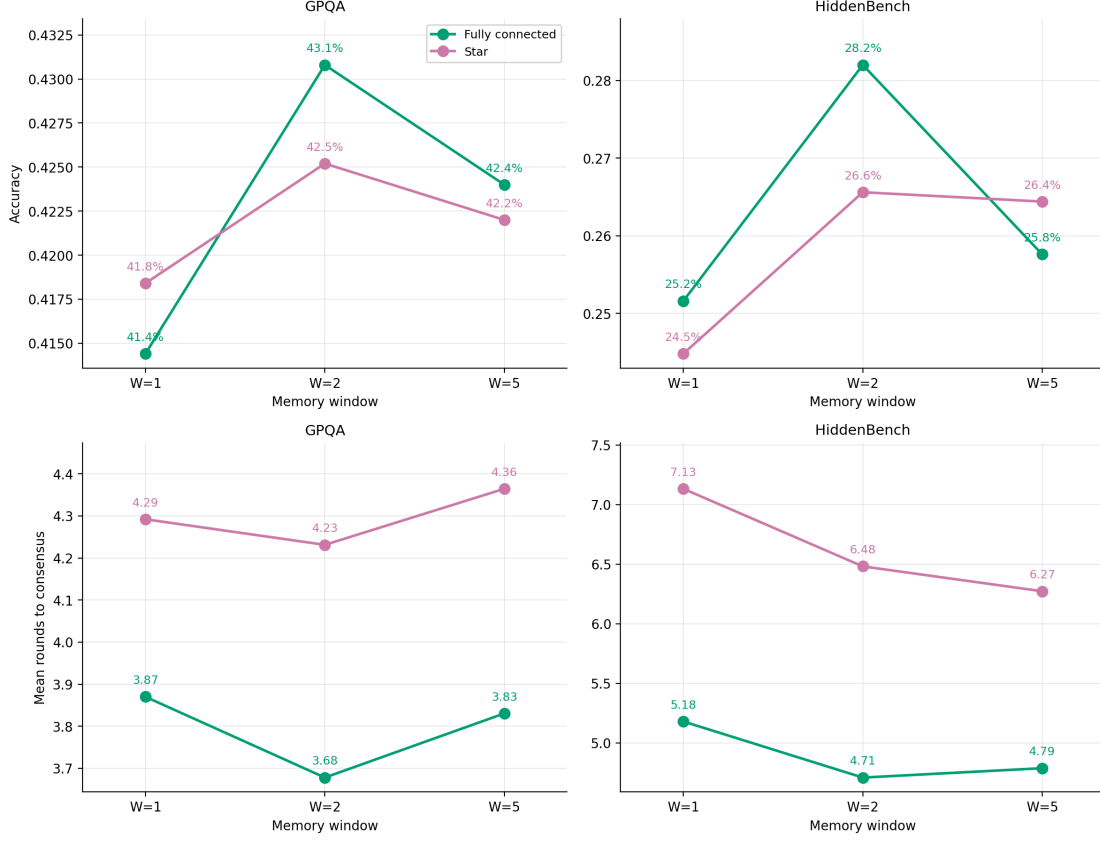


Figure 4: Accuracy in the top row and mean rounds to consensus in the bottom row as a function of memory window $W \in \{1, 2, 5\}$, for GPQA in the left column and HiddenBench in the right column. Each line is one topology, and the gap between the lines in the rounds panels is the topology efficiency effect. The y-axes do not start at zero, and the accuracy differences are not significant.

Neither structural axis has a significant effect on accuracy. Topology makes no detectable difference on either benchmark. Memory window shows a consistent pattern, as $W = 2$ yields the highest accuracy for both datasets and both topologies. However, the differences are small and do not survive multiple-comparison correction, so we read $W = 2$ as the best setting in our data rather than as an established effect.

Both axes do affect efficiency. The star topology needs more rounds to reach consensus on both benchmarks, and especially on HiddenBench, with $p < 0.001$ in both cases. The reason is that every spoke must route its messages through a single hub. The star partly compensates by sending half as many messages per round, so each round costs fewer tokens. Under the star topology, a larger memory window also speeds up convergence on HiddenBench. Exact values for all configurations, each with the standard error of

its accuracy across the 50 tasks, are given in Appendix A.4. This standard error stays below 0.05 in every configuration, so accuracy is reported as a point estimate without a dispersion term elsewhere in this thesis.

6.2 Self-Organization Motifs

Each motif is reported with a three-step template. Detection gives the prevalence against a null, drivers cover topology, W , and the initial vote composition, and outcomes relate the motif to accuracy and rounds to consensus.

6.2.1 Self-Reinforcement

Self-reinforcement asks whether an agent that keeps its vote grows steadily more confident in it. We read it from the slope β_{seg} of confidence against round number within each *vote-stable segment*, which is a run of at least three consecutive rounds on the same vote. We summarise a cell by the prevalence p_{SR} , the fraction of segments with a positive slope, against a random-walk null of 0.5. The full construction is given in Section 5.2. Pooled across both benchmarks and all configurations, the segment filter leaves 133,696 segments. Three findings follow.

- (i) **It is universal.** p_{SR} never approaches its null. It ranges from 0.733 to 0.883 across the 12 combinations of dataset, topology, and W , and all 50 tasks lie individually above 0.5 in every combination. Per-configuration values are given in Appendix A.5. Confidence climbs fastest where the debate has least to resolve, in repetitions that never change their votes and in unanimous starts. The initial vote composition drives the slope far more than topology or memory window do. Agents grow more certain in what they already held, not in what the debate revealed.
- (ii) **It drives convergence.** That rising confidence is what makes the debate settle. Tasks with steeper confidence slopes reach consensus in fewer rounds in all four combinations of dataset and topology. The task-level Spearman correlation lies between $\rho = -0.82$ and $\rho = -0.73$, as Figure 5 shows.
- (iii) **It is blind to correctness.** Faster convergence is not more accurate convergence. Whether self-reinforcement helps depends entirely on *what* it amplifies. When the correct option is already among the opening votes, stronger reinforcement pushes the group toward it. When no agent starts correct, the same force only locks in a wrong answer sooner. Appendix A.5 gives the corresponding accuracy analysis.

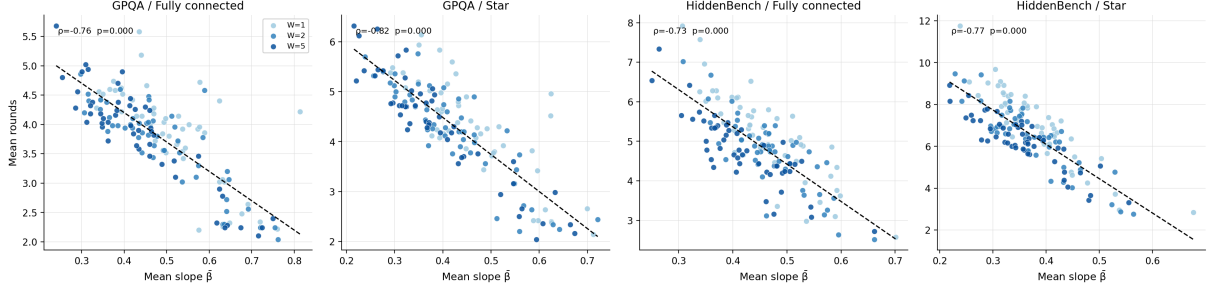


Figure 5: Mean segment slope $\bar{\beta}$ against mean rounds to consensus at the task level. Each panel is one combination of dataset and topology, each point is one task, and colours denote the memory window. The dashed line is the pooled OLS fit.

Self-reinforcement converges the debate, but it converges on whatever the opening votes already favour rather than on the correct answer. This matches the position lock-in that Oh et al. (2026) identify as a root cause of belief fixation, that Estornell and Liu (2024) derive from response homogeneity, and that Cau et al. (2026) describe as a selective amplifier of locally consistent change. It also mirrors reports that debate confidence rises without new evidence (Prasad and Nguyen, 2025, Bertalaníč and Fortuna, 2026). The seeded experiment in Section 6.3 confirms this dependence on the initial composition causally.

6.2.2 Multistability

Multistability asks whether a task admits more than one stable consensus outcome, and whether the opening vote distribution decides which one the debate reaches. We read it per task-configuration cell from two signatures. The first is N_{eff} , the inverse-Simpson diversity of the endpoint attractors, which counts how many stable outcomes coexist across repetitions. The second is Cramér’s V , which measures how strongly the initial votes select among these outcomes. A cell is *monostable* when one attractor dominates with $N_{\text{eff}} < 1.5$, *multistable* when several coexist and the initial votes predict the endpoint, and *stochastic* when several coexist but the endpoint does not follow from the start. The full construction is given in Section 5.3. Only repetitions that settle on a unanimous endpoint are included. Three findings follow.

- (i) **Most tasks are multistable, and the structure is set by the question.** Across the 600 task-configuration cells, 76 % show more than one coexisting attractor with $N_{\text{eff}} > 1.5$. This rate is conditioned on the selected task set. Neither topology nor memory window shifts either metric, and per-task N_{eff} is highly consistent across the six configurations. The structure is therefore a property of the question. Multiple absorbing states are theoretically expected here. Naming-game dynamics admit one equilibrium per available answer option (Baronchelli, 2018, Ashery et al., 2025), so a four-option task can in principle land on any of four attractors. Prior LLM debate studies could not observe this because they report a single run or run-level means

per task (Cau et al., 2026, Mehdizadeh and Hilbert, 2026). Our repeated-runs design makes the full attractor landscape visible.

- (ii) **Multistability helps exactly when the default answer is wrong.** Across tasks, more coexisting attractors go with *higher* accuracy on HiddenBench ($\rho = +0.44$) but slightly *lower*, non-significant accuracy on GPQA ($\rho = -0.16$). The opposition follows from what the dominant attractor is. On GPQA, where agents usually start near the correct answer, a single attractor is already the right one, and extra basins only add wrong-answer traps. On HiddenBench, where the majority attractor is the shared wrong answer, competing basins give the correct minority a foothold from which to win.
- (iii) **Fewer attractors produce stronger self-reinforcement.** Monostable cells carry the steepest confidence ramps, and N_{eff} anticorrelates with the mean self-reinforcement slope at the task level ($\rho = -0.74$ on GPQA and $\rho = -0.44$ on HiddenBench). When one attractor dominates, agents lock in early and confidence climbs unchecked. When several coexist, the reinforcement signal is split across diverging trajectories. Self-reinforcement is the within-basin amplifier, and multistability counts the basins.

Multistability is thus the structural counterpart of the amplifier. Each task carries a small set of stable outcomes fixed by the question itself. Which of those outcomes the group reaches depends on where it starts, as the causal analysis in Section 6.3 will show. A diversity of attractors therefore helps only when the single basin the group would otherwise collapse into is the wrong one. The per-dataset breakdown of monostable, multistable, and stochastic cells is given in Appendix A.6.

6.2.3 Dominance and Effective Topology

Dominance measures how unequally influence is shared among the agents. We compute it from the confidence-weighted conversion-credit graph as the Herfindahl index D . A value of 0 means that every agent converts equally many peers, and a value of 1 means that a single agent causes every opinion change. The full construction is given in Section 5.4. One structural caveat limits what we can test. In the star topology, a leaf can only ever be converted by the centre. Its raw D is therefore inflated by the wiring alone, and its surrogate null is degenerate for most repetitions. We report raw D for both topologies, but we test for *emergent* concentration only in the fully-connected topology, where all agents are structurally equal. We also describe the *shape* of the emerging hierarchy. For this, we map the conversion-credit graph of each fully-connected run to one of four structural types. In a *flat* run, influence is nearly uniform. A *hub* run has one moderately dominant agent, and a *two-hub* run has two agents of comparable influence. In a *star* run, a single agent is responsible for at least 75 % of all conversions. Figure 6 shows an example of each type. Four findings follow.

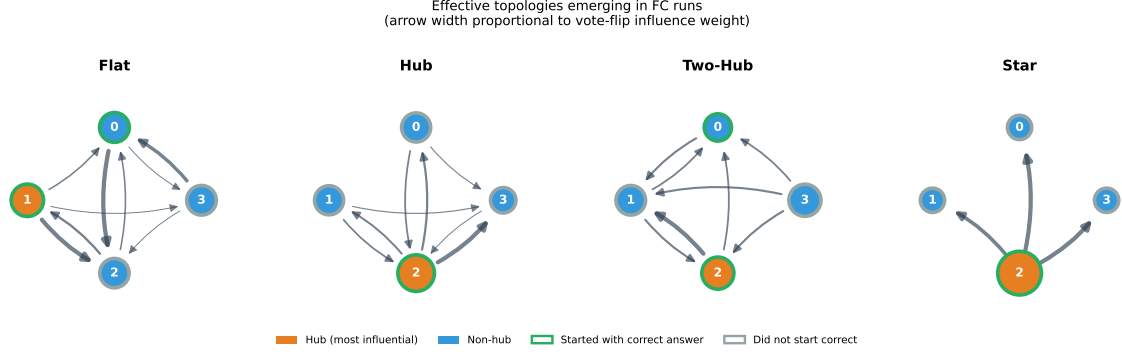


Figure 6: Examples of the four effective topologies observed in fully-connected runs with $N=4$. Arrow width is proportional to vote-flip influence weight. The orange node marks the hub, which is the most influential agent. Green borders mark agents that held the correct answer at round 0.

- (i) **A hierarchy emerges without a hub.** Every agent in a fully-connected run is wired identically, yet influence is almost never uniform. Conversion credit concentrates above the topology-keyed null in 49 of 50 GPQA tasks and in all 50 HiddenBench tasks. Flat runs are rare. By far the most common type is two-hub, in which two agents of comparable influence act as competing faction leaders. This type typically arises from an initial vote split of two against two. A clean star is rare, as Table 5 shows.
- (ii) **A leader helps only when the debate selects it.** When a star forms in a fully-connected debate, the group has chosen a hub that often started with the correct answer. On GPQA this holds for 54.5 % of star hubs and on HiddenBench for 85.5 %. A random agent is correct at round 0 only 38.2 % and 14.9 % of the time, as Table 4 shows. The group then almost always adopts the hub’s answer. If the hub started correct, about nine in ten runs end correct. If it started wrong, fewer than one in five do. This is why effective stars reach the highest accuracy of all types, with a clear margin on HiddenBench, as Table 5 shows. The picture reverses when the leader is imposed by the wiring. In the star topology on HiddenBench, the correlation between dominance and accuracy is negative at $\rho = -0.29$. The fixed centre is no more likely to be correct than any other agent, so it spreads the shared initial bias instead of correcting it.
- (iii) **Diverse initialisation is the lever that produces stars.** Whether a star forms depends strongly on how diverse the opening votes are. Stars almost never form when all agents start on the same option. When all four agents start on different options, roughly one in five runs becomes a star, as Figure 7 shows. Enforced perspective diversity has already been proposed as a remedy for belief lock-in (Oh et al., 2026), and it also follows from the theoretical analysis of debate homogeneity (Estornell and Liu, 2024). Our contribution is the mechanism behind it. Diversity raises the chance that the debate elects an informed hub.

- (iv) **Concentration costs rounds.** Whatever its effect on accuracy, dominance consistently slows convergence. Within each task, configurations with a higher mean D need more rounds to reach consensus. The per-task Spearman correlation is $\rho = +0.60$ on GPQA and $\rho = +0.66$ on HiddenBench. A dominant agent has to press its position over many rounds before the other agents give in.

Effective topology	Share of FC runs		Accuracy	
	GPQA	HiddenBench	GPQA	HiddenBench
Flat	5.4 %	6.8 %	0.394	0.242
Hub	11.0 %	18.0 %	0.393	0.377
Two-hub	58.7 %	59.2 %	0.418	0.247
Star	2.3 %	5.4 %	0.485	0.562
Unclassified	22.6 %	10.6 %	0.450	0.029
All FC runs			0.423	0.264

Table 5: Prevalence and accuracy of the four effective topology types in fully-connected runs, with 7,500 runs per dataset. Unclassified runs contain no vote flips and therefore have no conversion-credit graph.

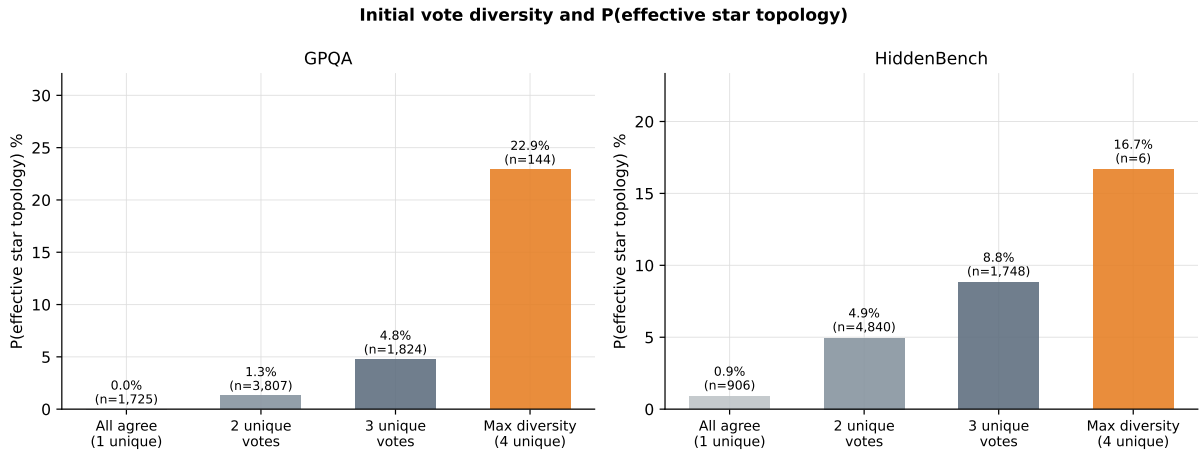


Figure 7: Probability of an effective star topology as a function of initial vote diversity. n_{unique} is the number of distinct votes held at round 0. The maximum-diversity bin contains 144 GPQA runs but only 6 HiddenBench runs.

Together, dominance and effective topology show the amplifier from two complementary angles. The debate reliably elects an influence hierarchy that the wiring never specified, and the group’s answer follows whichever agent wins that election. Accuracy is high when the elected hub started correct, and it suffers when a fixed hub entrenches a shared error. This is the structural counterpart to the causal test in Section 6.3, which shows the same dependence on the opening votes by direct manipulation. It also mirrors reports that nominally symmetric LLM populations develop structural asymmetry on

their own, both through preferential-attachment links (Mehdizadeh and Hilbert, 2025a) and through locally clustered agents coordinating better than high-betweenness bridges (Mehdizadeh and Hilbert, 2026). Per-cell concentration values across all configurations are in Appendix A.7.

6.2.4 Limit Cycles

Limit cycles are the rarest motif in this system, and the only one that is a failure rather than a variant of convergence. A limit cycle is a debate that never settles and oscillates between vote configurations instead of reaching a fixed point. We detect them with categorical recurrence quantification. A sequence is flagged when its determinism exceeds a shuffle-null floor and its dominant period lies in a genuine oscillatory band with $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$. The full construction is given in Section 5.5. The cooperative design suppresses oscillation almost entirely. We test at two levels, the vote sequence of each individual agent and the vote composition of the whole group per repetition. At the agent level, 99.5 % of sequences converge to a fixed point, and at the system level 98.7 % do. These converged sequences never enter the test. Three findings follow.

- (i) **They are rare but genuine.** Within the small non-converged remainder, 47 agent-level and 32 system-level sequences are flagged, both far above the shuffle-null floor. On inspection, 70 % of the flagged agent-level cases are clean period-2 orbits whose recurrence matrices show the characteristic checkerboard pattern. The remaining cases are near-convergent artefacts.
- (ii) **They occur only in the star topology.** Almost every flagged case sits in a star configuration, 45 of 47 at the agent level and 26 of 32 at the system level. The few flagged fully-connected cases all turn out to be stable revisits or near-convergent artefacts on inspection. Flagged cases also concentrate on HiddenBench, with 34 of 47 at the agent level. Figure 8 shows the two clearest examples.
- (iii) **The mechanism is hub-driven majority inversion, purely an efficiency failure.** Every limit cycle runs to the hard round cap without ever reaching consensus, so early stopping never fires. Inspection suggests a consistent, though uncertified, causal chain. Agents defer to the peer majority they observe. After each flip, the star hub rebroadcasts that majority for the opposite option. With two independently defensible lines of evidence and a hub that starts in the minority, each flip recreates the conditions that caused it. This also explains why the fully-connected topology shows no genuine cycles. Without a single node that aggregates all peer signals, the inversion loop cannot form. In practice, repetitions that reach the cap should therefore be flagged as unresolved rather than reported as a decision. Failure to recognise termination conditions is a catalogued failure mode of deployed multi-agent systems (Cemri et al., 2026), and our result names a specific dynamical route to it.

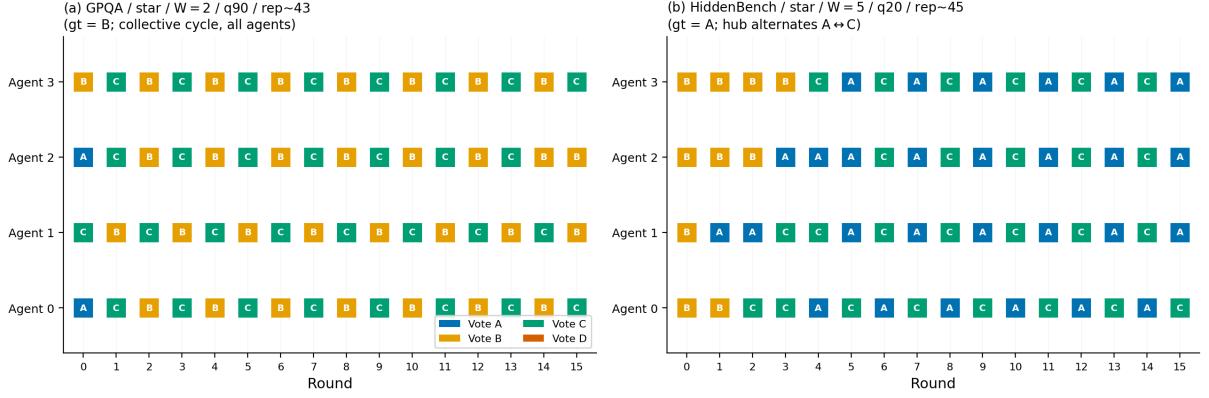


Figure 8: Vote-sequence timelines for two limit cycles in the star topology. Each cell shows the vote held by an agent in a given round, and colour encodes the option. Both panels show the same hub-driven majority inversion, in which three agents alternate in phase and one agent alternates in anti-phase. In (a), from GPQA with $W=2$ and ground truth B, the cycle starts almost immediately. In (b), from HiddenBench with $W=5$ and ground truth A, the group first agrees on B and only falls into the cycle from round 5 onward. Neither repetition reaches consensus, and both run to $T = 15$.

Limit cycles are therefore the failure mode of the same convergence machinery seen elsewhere. When a single aggregator keeps reconstructing the opposite majority, self-reinforcement never gets to lock the group in, and the debate oscillates instead of settling. That this happens only under the star, and never when every agent sees every other, fits the observation that unanimous states are otherwise effectively absorbing (Li et al., 2024). The full detection funnel and representative recurrence matrices are given in Appendix A.8.

Motif summary. Table 6 collects the four motifs side by side. Three of them point back to the opening vote composition. Self-reinforcement amplifies it, multistability lets it select the basin, and the formation of an elected hub depends on its diversity. Limit cycles are the exception, since they arise from the star wiring rather than from the opening votes. Section 6.3 then manipulates the opening vote composition directly.

Motif	Detection	Main driver	Accuracy	Rounds
Self-reinforcement	In every cell and task	Opening votes	Helps only if the correct answer is present at the start	Fewer
Multistability	In most cells	The task itself	Higher on HiddenBench, no significant link on GPQA	Not tested
Dominance and effective topology	Above the null in fully-connected runs	Diversity of the opening votes	Helps if the hub is elected, hurts if it is imposed	More
Limit cycles	Rare, star topology only	Star hub	None, the debate stays unresolved	Always hits the cap

Table 6: Summary of the four motif findings. Detection refers to prevalence against the respective null, and accuracy and rounds describe the direction of the association.

6.3 Causal Test of the Initial-Condition Effect

The motif analyses point to the opening vote composition as the factor that decides which attractor the debate reaches. This association is confounded, because whether the ground truth is present at $t = 0$ co-varies with task difficulty, agent confidence, and initial vote entropy. Three experiments address this in sequence. A within-task observational analysis at $R = 1,000$ establishes the pattern. A seeded dose-response fixes the opening composition directly and tests the causal interpretation. An adversarial probe then stress-tests the robustness of the amplifier.

Ground-truth presence in the initial vote distribution. For the observational analysis, three GPQA questions, q84, q125, and q144, were each run for $R = 1,000$ repetitions at $W = 2$ in the fully-connected topology with $N = 4$. They were selected for mid-range accuracy and high initial-vote entropy, so that their outcomes are genuinely uncertain. The dominant finding is striking. Whether the ground-truth option appears at all in the initial vote distribution is by far the strongest predictor of final accuracy, consistently across all three tasks, as Figure 9 and Table 7 show.

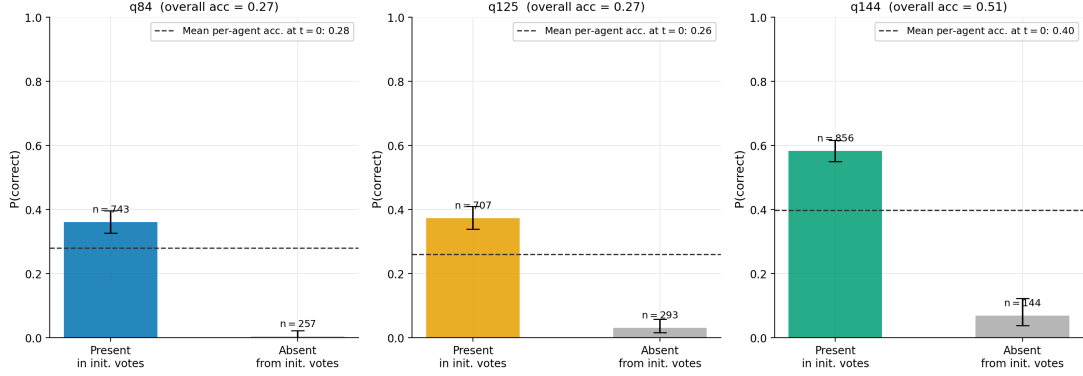


Figure 9: System accuracy conditioned on whether the ground-truth option is present in the initial vote distribution at $t = 0$. Left bars show repetitions in which at least one agent voted correctly before debate, and right bars show repetitions in which no agent did. The dashed line marks the mean per-agent accuracy at $t = 0$. Error bars are Wilson 95 % confidence intervals (Wilson, 1927). The data cover three GPQA tasks with $R = 1,000$ each, $W = 2$, and the fully-connected topology.

Task	Overall acc.	GT absent (acc., n)	GT present (acc., n)	GT is plurality (acc., n)
q84	0.269	0.004 (257)	0.361 (743)	0.620 (213)
q125	0.273	0.031 (293)	0.373 (707)	0.552 (250)
q144	0.509	0.069 (144)	0.583 (856)	0.717 (403)

Table 7: Accuracy by ground-truth presence in the initial vote distribution, with $R = 1,000$ per task. GT absent and GT present indicate whether the correct option appears in the vote profile at $t = 0$. GT is plurality is the subset of present repetitions in which the correct option holds the initial plurality.

When the correct answer is absent from all initial votes, the system almost never reaches it. Accuracy is 0.4 %, 3.1 %, and 6.9 % for q84, q125, and q144, with $\chi^2 > 120$ and $p < 10^{-27}$ in all cases. The rare correct answers in the absent condition are consistent with stochastic noise rather than genuine recovery. Within the present condition, accuracy is roughly twice as high when the correct option also holds the initial plurality. For q84, for example, accuracy is 0.620 when the correct option leads the opening votes and 0.257 when it is present but does not lead. These findings suggest that the system operates primarily as a vote amplifier. Debate selectively strengthens options already represented in the initial distribution and overwhelmingly reinforces whichever option holds the initial plurality. The capacity to surface a correct answer that no agent endorsed at the outset is negligible under the present design.

Neither the length of a debate nor its confidence trajectory predicts whether it is right. Longer debates are no more accurate, and confidence rises monotonically across rounds whether the final answer is correct or not. Appendix A.9 gives the full feature-by-outcome

correlations. What the outcome does track is the opening vote composition, which the next experiment manipulates directly.

Seeded dose-response. To isolate the causal contribution of the opening composition, we fix it directly. We use 50 GPQA tasks in the same configuration with $W = 2$, the fully-connected topology, $N = 4$, and $R = 50$ repetitions each. For every repetition, we build the initial four-agent vote profile by sampling from a pool of genuine, independent single-agent responses under a fixed number of correct seeds $k \in \{0, 1, 2, 3\}$. Exactly k of the four agents begin on the correct option, and the rest begin on real wrong answers. Only the initial composition is manipulated. Every vote is a real response, and the debate that follows is unmodified. Sweeping k turns the earlier binary contrast into a dose-response curve.

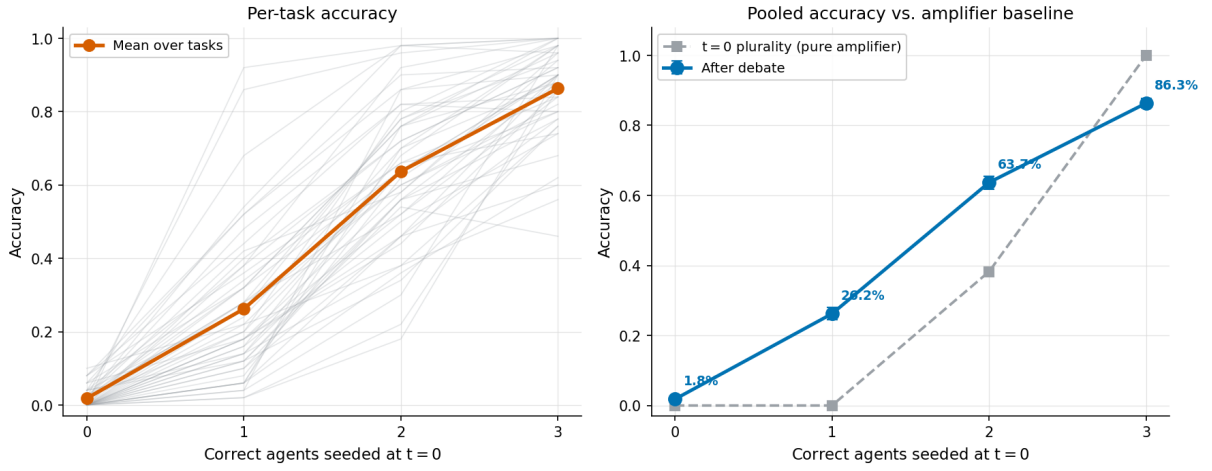


Figure 10: Accuracy as a function of the number of correct agents seeded into the initial vote profile, $k \in \{0, 1, 2, 3\}$, on 50 GPQA tasks with $R = 50$ each, $W = 2$, and the fully-connected topology. Panel (a) shows per-task accuracy in grey, one line per task, and the mean over tasks in orange. Panel (b) shows pooled accuracy in blue against the accuracy that a pure plurality vote at $t = 0$ would already achieve, shown as a dashed grey line. Debate exceeds this amplifier baseline for a correct minority or tie but falls below it for a correct supermajority.

Three findings follow.

- (i) **Accuracy rises monotonically with the number of correct seeds.** Pooled over 2,500 repetitions per condition, accuracy climbs from 1.8% at $k = 0$ to 26.2%, 63.7%, and 86.3% at $k = 1, 2, 3$, as Figure 10 shows. This is not merely a pooled average. Adding one correct seed raises per-task accuracy in 50 of 50 tasks for each of the first two steps and in 47 of 50 for the last. Causally, the composition of the opening votes is the dominant lever on the outcome.
- (ii) **Debate is a lossy amplifier, not a reasoner.** Comparing each condition against the accuracy that a pure plurality vote at $t = 0$ would already achieve isolates

what the debate itself contributes. This baseline is the dashed line in Figure 10. Where the correct answer starts behind, debate helps. A lone correct minority at $k = 1$, where plurality accuracy is 0%, is carried to the right answer 26% of the time. An even split of two against two at $k = 2$, with a baseline of 38%, resolves correctly 64% of the time. Where the correct answer starts ahead, debate *erodes* it. A correct supermajority of three at $k = 3$, which plurality voting would preserve with certainty, is talked out of the right answer in roughly one repetition in seven, reaching 86% rather than 100%. From a standing start, debate generates almost nothing. It recovers the answer de novo in only 1.8% of $k = 0$ repetitions, confined to 21 of 50 tasks. Debate amplifies the initial signal while blurring it toward a coin flip, rather than reasoning its way to the truth.

- (iii) **Outcomes are all-or-nothing and decided early.** Every repetition in every condition ends fully unanimous. The correct seeds either convert the whole group or capitulate entirely, and no repetition ends with the ground truth as a surviving minority. When the correct side wins, it wins fast and reaches a plurality within the first one to two rounds, with a mean first round of 1.3. Persuasion is front-loaded. The opening exchange, not protracted deliberation, decides the outcome.

Adversarial probe. The amplifier does not care whether it amplifies a correct or a wrong answer, and this becomes most visible when one agent acts as an adversary. We re-run the $k = 2$ condition with $R = 25$, but now one of the two agents without the correct answer is told to keep arguing for a wrong option throughout the debate. Accuracy drops from 63.7% to 10.0%, and it drops in every single one of the 50 tasks. The group now does worse than a single correct agent without any adversary, which still reaches 26.2% at $k = 1$. This happens even though the adversary starts with two correct agents against it. Most of these runs still reach unanimity, and when they do, the group almost always ends on the adversary’s wrong answer. The reason is simple. An agent that never changes its mind looks convinced, and the debate rewards conviction. The same mechanism that sometimes lets a correct minority win now works for the adversary, so the group follows the most persistent voice rather than the most correct one. This matches the tipping-point dynamics that Centola et al. (2018) report for human conventions, where a committed minority of about 25% is enough to overturn the majority. Our adversary is exactly such a committed minority of one in four agents. Similar effects have been reported for LLMs. Agarwal and Khanna (2025) show that a forcefully argued falsehood can override a truthful answer in an LLM judge’s decision, and Yao, Shang, Du, He, Lian, Zhang, Su, Swamy and Qi (2025) describe how agents give in to each other too early, a problem that prompt-based fixes now try to address (Pitre et al., 2025).

Taken together, the three experiments upgrade the amplifier reading from correlation to intervention. The multi-agent system is a *lossy amplifier of its initial vote distribution*. It reliably strengthens whatever the opening votes favour, more so as that signal grows, yet it neither manufactures a correct answer that no agent held nor protects a correct one that a persistent dissenter attacks. The composition of the opening votes, not the length or intensity of the debate, is the dominant lever on whether collective reasoning succeeds. This intervention was run on GPQA only. The observational ground-truth-presence result

holds on both benchmarks, as Section 6.2.1 shows, and the extension of the causal claim to HiddenBench is discussed in Section 7.4.

7 Discussion: From Dynamics to Diagnostics

Section 6 reported what the system does. This section asks what it means, in four steps. Section 7.1 explains *why* the system behaves as it does by locating it in the vocabulary of Section 2.1. Section 7.2 asks what that behaviour *costs*, by measuring the debate against the vote it started from. Section 7.3 asks what a practitioner can *do* about it. Section 7.4 states the bounds on all three together with the work that would remove them.

7.1 The Missing Ingredient

This subsection answers why the system behaves as it does, and it does so by returning to the definition of self-organization rather than by adding new measurements. Previously, in Section 2.1 introduced the four ingredients that Camazine and colleagues identify as recurring in self-organizing systems, which are positive feedback, negative feedback, fluctuations, and repeated local interaction. Three of the four are clearly present in our results. Positive feedback is self-reinforcement, which appears in every one of the twelve combinations of dataset, topology, and memory window at a prevalence between 0.733 and 0.883 against a null of 0.5, and which is the engine that makes the debate settle, since tasks with steeper confidence slopes reach consensus in fewer rounds in every combination we tested. Fluctuations are visible as multistability, since 76 % of the 600 task and configuration cells admit more than one stable outcome across repetitions of the same question. Repeated local interaction is the debate protocol itself. Thus, the second ingredient, negative feedback, is the one that is missing. Nothing in the protocol bounds an early lead. No agent is rewarded for holding out, the group pays no cost for converging quickly, and confidence climbs fastest in exactly those repetitions where the debate has least to resolve, which are the ones that never change their votes and the ones that start unanimous.

That absence is the finding. A system with amplification and no damping does not deliberate its way to an answer, it magnifies whatever asymmetry it starts with. This also shows that SOMA divides the problem in a useful way, because the four detectors turn out to measure not four phenomena but one mechanism seen from four angles. Self-reinforcement is the view within a stable segment, where an agent grows more certain of a position it already held rather than of anything the exchange revealed. Dominance is the view within a run, where influence concentrates above the topology-keyed null in all but one of the hundred task and benchmark combinations even though every agent in a fully connected run is wired identically, so the group reliably elects a hierarchy that the structure never specified. Multistability is the view across repetitions, where the same question lands in different basins on different runs because the opening draw differs, and it counts the basins that self-reinforcement then collapses into one. Limit cycles are the boundary case where the magnification never terminates, and they are informative precisely because they are almost absent, since nearly every sequence reaches a fixed point and the 47 agent level and 32 system level flagged cases sit almost entirely in the star topology. That the fully connected topology produces no genuine cycles is a clean

statement about cooperative protocol design rather than a measurement failure, because without a single node that aggregates all peer signals there is no way to reconstruct the opposing majority after every flip.

Read this way the four motifs yield one claim rather than four findings. Within the system we tested, deliberation works predominantly as a selective amplifier of its initial vote distribution rather than as a mechanism that reasons a new answer into existence. The claim is not novel in kind. Oh et al. (2026) name homogenised dynamics that amplify the majority view regardless of correctness as a root cause of belief entrenchment, and Estornell and Liu (2024) derive the same conclusion theoretically from response homogeneity. What SOMA adds is measurement. Each face of the mechanism is separately detectable from ordinary interaction logs and each was tested against a null that destroys the pattern and leaves everything else intact, so a practitioner can check which regime a system is in rather than having to assume it.

7.2 A Lossy Amplifier

This subsection asks what the missing ingredient costs, by measuring the debate against the only baseline that isolates its contribution, which is the vote the same four agents already held before talking. The seeded experiment answers that directly, because it fixes the number of correct agents in the opening profile and leaves the debate itself untouched. Accuracy climbs from 1.8% with no correct seed to 26.2%, 63.7%, and 86.3% as one, two, and three agents begin on the correct option, and it does so in almost every individual task. Set against what a plurality vote at round 0 would already deliver, the pattern is not a uniform gain. Where the correct answer starts behind, the debate helps, since a lone correct agent faces a plurality baseline of zero and still wins a quarter of the time, and an even split of two against two improves on a baseline of 38%. Where the correct answer starts ahead, the debate erodes it, since a correct supermajority of three that plurality voting would preserve with certainty is talked out of the right answer in roughly one repetition in seven.

Debate is therefore better described not as majority voting with reasoning attached but as a transformation that widens the distribution around the opening signal. It moves out-comes away from the initial plurality in both directions, which is useful exactly when that plurality is unreliable and harmful when it is already good. This one property accounts for the two operating regimes we observe and, we think, for much of the disagreement in the literature about whether debate works. On GPQA the agents usually begin with the correct option present, the plurality of the opening votes already reaches 40.5%, there is little room above that baseline, and debate adds under two percentage points. On HiddenBench the shared model bias drives the opening plurality below chance, because the errors are correlated in the way that biased information sampling produces in human hidden profile groups (Stasser and Titus, 1985, Li et al., 2026), and the same widening recovers 18.5 percentage points. The two benchmarks are not an easy and a hard version of one task, they are the two sides of the crossing point where the transformation stops paying, observed inside a single experimental setup. Results reported on benchmarks that

sit on one side or the other will disagree about debate for reasons that have nothing to do with the protocol, which is consistent with the finding that debate often fails to beat simple baselines (Zhang et al., 2025) and that majority voting alone accounts for much of the apparent gain (Choi et al., 2025).

The motif signatures track the same crossing point, which is the strongest internal evidence that one mechanism is at work. Multistability helps on HiddenBench, where more coexisting attractors go with higher accuracy at $\rho = +0.44$, because a competing basin gives the correct minority a foothold against a dominant attractor that is wrong, while on GPQA the relationship is weakly negative and not significant, which is what we would expect where the single dominant basin is already the right one. Dominance reverses in the same way. When the debate elects its own hub in a fully connected run, that hub started correct far more often than a random agent does, and effective star runs reach the highest accuracy of the four structural types on both benchmarks. When the hub is imposed by the wiring instead, the fixed centre is no more likely to be correct than anyone else, so it spreads the shared bias rather than correcting it, and higher concentration then goes with lower accuracy.

The adversarial probe shows what the transformation actually tracks, and it is not correctness. Replacing one of the two incorrect agents in the two seed condition with an agent that keeps arguing for a wrong option throughout collapses accuracy from 63.7% to 10.0%, in every task, and leaves the group below what a single correct agent with no adversary achieves. Two correct agents out of four are not enough to hold the line against one that never moves. The amplifier responds to persistence, and since a cooperative agent that is right and an adversarial agent that is committed produce the same observable signal, the protocol has no basis on which to tell them apart. Whatever lift truth seeding provides is equally available to anyone willing to supply a confident wrong answer instead.

7.3 What to Measure Instead

The motivation for this thesis was that outcome metrics tell a practitioner whether a configuration worked but never why, and give no way to separate a healthy deliberation from a pathological one (Miehling et al., 2025). The negative half of our answer is that the signals an operator would naturally reach for instead are uninformative or inverted. Confidence rises monotonically across rounds whether the final answer is correct or not, so a confident group is not a reliable one. Unanimity is worse than uninformative on HiddenBench, where repetitions that stop early lock in the shared wrong answer almost universally and reach only 2.5%, because early consensus forecloses the exchange that the benchmark requires. Debate length carries no signal either, since longer debates are no more accurate. The same holds for the two structural knobs the literature most often tunes (Li et al., 2024, Mehdizadeh and Hilbert, 2026). Neither topology nor memory window has an accuracy effect that survives correction for multiple comparisons, and both are clearly visible in the dynamics, since the star needs more rounds on both benchmarks, is the sole source of genuine limit cycles, and concentrates influence by construction. Network type

having no main effect on coordination in an eight topology factorial (Mehdizadeh and Hilbert, 2026) and communication density failing to lift accuracy above a non communicative baseline (Bertalaníč and Fortuna, 2026) point the same way, and the dissociation explains why topology tuning results in the literature are mixed.

The positive half is that the composition of the opening votes is readable, cheap, and decisive, and that one design choice does reach it. The effective influence topology that emerges inside a fully connected run predicts accuracy even though the prescribed topology does not, because the debate elects a hub that disproportionately started correct while a prescribed hub is fixed before the debate can select anyone. Whether that election happens depends on how diverse the opening votes are, since stars almost never form when all four agents start on the same option and roughly one run in five becomes a star when all four start on different ones. Diverse initialisation is therefore the one lever in our grid that touches accuracy, and it does so because it acts on the input to the amplifier rather than on the amplifier itself. Enforced perspective diversity has already been proposed as a remedy for entrenchment (Oh et al., 2026), and what we add is the mechanism behind it.

Table 8 states the resulting guidance together with the result it rests on, the strength of that evidence, and the quantity that checks it. We do not deliver a validated diagnostic tool, and the table should be read as a set of hypotheses with their provenance attached rather than as a specification. What it does show is that every recommendation here is checkable from ordinary interaction logs using a SOMA detector that has already been tested against a null model on 30,000 repetitions, which is what we mean by groundwork for a diagnostic suite rather than the suite itself. Taken together the guidance is deflationary, and usefully so. A system whose answer is largely fixed before deliberation begins, that cannot manufacture a correct answer absent from its inputs, and whose own confidence does not track correctness is one whose reliability has to be established upstream in the agents that produce the opening positions and externally through verification that does not trust the group, rather than through longer or more elaborate debate.

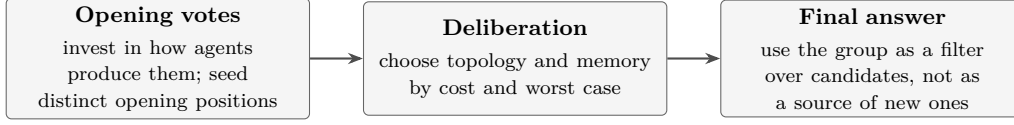
Guidance	Evidence	Observable	How a builder checks it
Invest in the process that generates the opening votes rather than in tuning the debate	Causal, seeded dose response (Section 6.3)	$P(\text{GT present})$, opening plurality	Log the opening vote profile and relate it to the outcome across repetitions
Use the system as a filter over candidate answers, not as a way to discover one no agent holds	Causal, the zero seed arm (Section 6.3)	De novo recovery rate	Measure the fraction of repetitions whose final answer was absent at $t = 0$
Do not gate trust on the group’s confidence, its unanimity, or the length of the debate	Every cell, self-reinforcement and early stop analyses (Section 6.2.1)	Confidence gain by outcome, accuracy of early stops	Check confidence calibration against held out truth
Seed distinct opening positions on tasks with four or more options	Observational, effective topology analysis	Opening vote diversity, effective topology type	Track how often an effective star forms as opening diversity rises
Treat topology and memory as efficiency levers and flag capped repetitions as unresolved	Every cell, the accuracy and efficiency grid and the limit cycle detector	Rounds, cap rate, dominance D	Compare cost and worst case across topologies rather than accuracy
Treat an outcome that follows a single agent that never updates its vote as that agent’s answer	Single probe, the adversarial arm (Section 6.3), and the effective topology analysis	Vote entropy per agent across rounds	Flag near zero vote entropy and verify that agent’s answer externally

Table 8: Design guidance, the result it rests on, the strength of that evidence, and the measurement that checks it. Every observable is computed in this thesis from ordinary interaction logs. The suite itself is not built here, this is the set of instruments it would use.

Figure 11 orders the same guidance by when a practitioner encounters it. Panel (a) collects the choices made while building the system: investing in the opening votes, choosing topology and memory by cost and worst case, and using the group to select among candidate answers rather than to generate new ones. Panel (b) turns the remaining observables into checks on the log of a single debate. A run that reaches the round cap without converging is unresolved, a final answer that no agent held at $t = 0$ lies outside the amplifier’s normal behaviour, and an outcome that follows a single agent that never changed its vote is only as reliable as that agent. A debate that passes all three checks is accepted only pro-

visionally, because neither its confidence, its unanimity, nor its length indicates whether it is correct. Like the table, the figure states hypotheses grounded in our results rather than a validated procedure.

(a) Designing the system



(b) Auditing one debate from its log

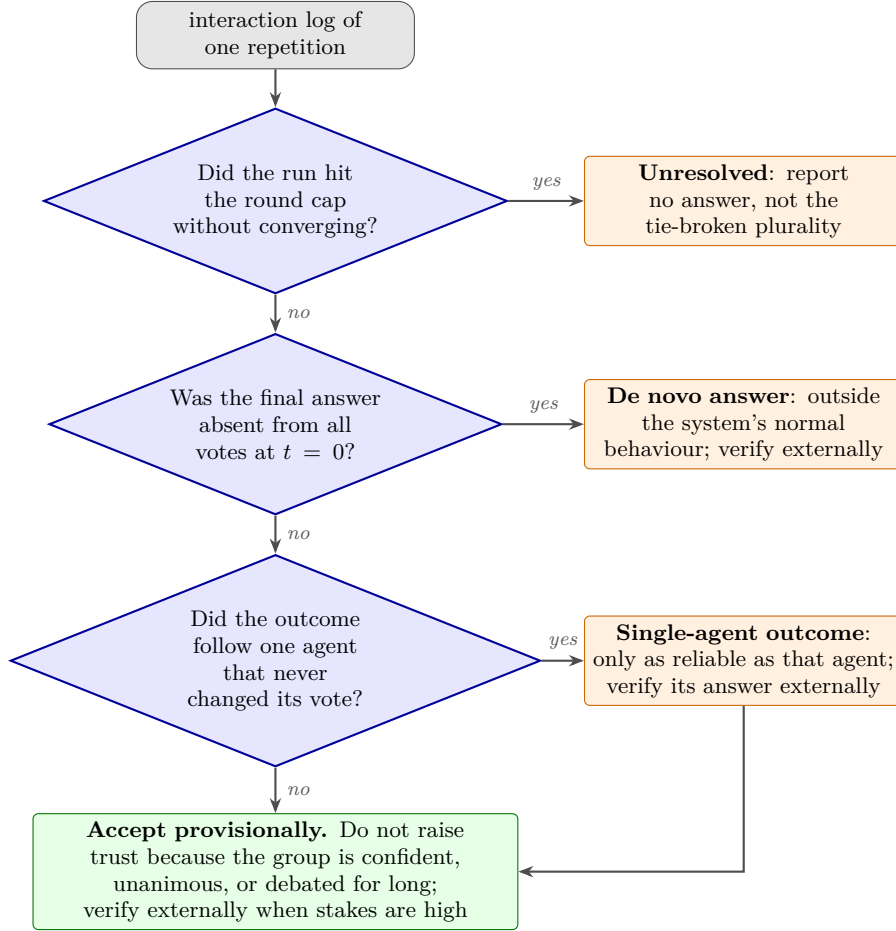


Figure 11: The design guidance of Table 8 ordered into a decision procedure. **(a)** Choices made when building the system, from the opening votes to the final answer. **(b)** Checks applied to the log of a single debate before its answer is used. The procedure holds within the tested setting (one model, four agents, two benchmarks).

7.4 Limitations and Future Work

The first limitation is the scope of the experimental grid. We varied communication topology, memory window, task difficulty, and benchmark, and in the seeded experiment the composition of the opening votes, but we did not vary the base model, the group size, the homogeneity of the agents, or the cooperative framing. Every result reported here is therefore a statement about a homogeneous, cooperative, four agent debate on one model family, and nothing in it establishes what happens outside that setting. Each of these fixed choices is a direction for further work. Group size is the most immediate, since $N = 4$ with three or four answer options leaves a state space small enough that votes repeat and influence concentrates by chance alone, which is why every detector required a null in the first place, and a larger group would relax that constraint while letting the effective topology analysis distinguish structures that four agents cannot express. The structural knobs are narrow for a related reason, since fully connected and star are the two extremes of a much larger space, and intermediate structures such as rings, sparse or layered graphs, and dynamically rewiring ones would show whether hub driven cycles are specific to a single aggregator or arise wherever information is funnelled through a bottleneck. The memory window we tested is short relative to the round cap, and at a sufficiently large W a spoke in a star can reconstruct the whole exchange from accumulated hub summaries, at which point the structural difference between the topologies largely disappears, so extending W upward would locate the point at which the distinction stops mattering. Agent heterogeneity is the axis the critical literature considers decisive, since it identifies model diversity rather than debate itself as the reliable source of gains (Zhang et al., 2025, Chen, Saha and Bansal, 2024) and capability asymmetry as a driver of harmful herding (Wynn et al., 2025).

The second limitation is the reach of the causal evidence. The seeded experiment is our only intervention and it was run on GPQA, so the amplifier claim is causal there and observational on HiddenBench, where it rests on the association between the presence of the ground truth in the opening votes and the final answer. The diverse initialisation lever is observational throughout, because we measure that the probability of an effective star rises with opening diversity across naturally occurring runs rather than manipulating that diversity ourselves, and the maximum diversity condition holds only six runs on HiddenBench. The effective topology analysis is further confined to fully connected runs, since a leaf in a star can only ever be converted by the centre, and its four types are separated by thresholds on the share of influence captured by the top one or two agents, so the rare types at 2.3 % and 5.4 % of runs carry real uncertainty. Two interventions would address this. Seeding each agent with a distinct opening answer would test the diversity lever directly, and perturbing or removing the elected hub would establish whether its influence is causal or whether it proxies for question specific expertise.

The third limitation is what SOMA is able to certify. All four detectors identify signatures rather than dynamical objects. A positive confidence slope is not a proof of a feedback loop, a recurrence test flags periodic structure rather than a mathematically certified limit cycle, N_{eff} counts coexisting endpoints rather than a certified attractor landscape, and the dominance index records where conversions concentrate rather than who caused

them. Two of the detectors also have a known direction of error. The self-reinforcement prevalence may be inflated, because an agent whose confidence falls is more likely to flip its vote and end the segment, which censors negative slopes, and a matched null that models the flip hazard would show how much of the excess above the baseline is genuine. The dominance graph is built from vote adoptions during the *Update* phase, which is precisely where the protocol instructs agents to update, so a strong hub to spoke edge may reflect persuasion or merely the routing of information. Removal based attribution, which estimates an agent’s contribution by masking its messages (Lu et al., 2026, Cui et al., 2025), would replace the concentration index with an interventional measure of influence and is the natural successor to what we report here.

The fourth limitation concerns what the reported rates describe. The tasks were screened for outcome variation across repetitions, so the prevalences we report characterise the regime in which the outcome is genuinely contested rather than the full distribution of questions. This follows from the objective, which was to characterise the dynamics rather than to evaluate the benchmarks, but it does mean the rates should not be read as how often each motif would appear in arbitrary deployment, and establishing that would require running the unscreened task set. The study is also exploratory and reports many tests across twelve configuration cells and four motifs, so we have leaned throughout on effect sizes and on consistency across cells rather than on isolated significance, and the central claims rest on effects that are large and repeated rather than on any single borderline test.

8 Conclusion

This thesis asked whether the dynamical vocabulary of complex systems can be applied empirically to LLM-based multi-agent debate, and what such a treatment reveals that outcome metrics cannot. To answer it we built a two phase debate protocol whose logs expose private beliefs and confidences alongside public messages, defined SOMA as four motif detectors each grounded in the complex systems or opinion dynamics literature and each tested against its own null model, and ran the full grid on two deliberately contrasting benchmarks across 30,000 debates.

The first research question asked whether the motifs can be measured at all. SOMA detects them, at very different prevalences, and the differences are themselves informative. Self-reinforcement is universal, with a prevalence between 0.733 and 0.883 in every combination of dataset, topology, and memory window. Multistability is common and belongs to the question rather than to the configuration, since 76 % of the 600 task and configuration cells admit more than one stable outcome and per-task values are consistent across configurations. Dominance emerges behaviourally even where the wiring is symmetric, concentrating above the topology-keyed null in all but one of the hundred task and benchmark combinations. Limit cycles are rare and real, since almost every sequence reaches a fixed point and the small residue yields 47 agent level and 32 system level flagged cases, most of them clean period two orbits on inspection. That they sit almost entirely in the star topology, and that the fully connected topology produces none, is a clean result about cooperative protocol design rather than a measurement failure.

The second research question asked what governs the motifs and what they predict. The two structural axes shape the dynamics and not the correctness of the debate. The star topology is slower on both benchmarks, concentrates influence by construction, and is the only source of genuine cycles, while the memory window affects convergence speed only and multistability is invariant to both. What governs the outcome instead is the composition of the opening votes. Self-reinforcement is the convergence mechanism everywhere, yet it strengthens whichever option leads at the outset rather than the correct one, so faster convergence is not more accurate convergence. Multistability decides how many outcomes are available to be selected and helps only where the single basin the group would otherwise fall into is the wrong one. Dominance helps when the debate elects its own hub, because an elected hub started correct far more often than a random agent, and hurts when the hub is prescribed by the wiring, because a fixed centre spreads the shared bias instead of correcting it. Limit cycles mark pure efficiency failures, since every one of them ran to the cap without producing a decision.

The third research question asked whether the opening votes are causally responsible for the outcome. They are. Observationally, whether the ground truth appears at all in the opening distribution is the strongest predictor of accuracy we measured. Causally, seeding one correct agent into an otherwise wrong opening raises accuracy from 1.8 % to 26.2 % and the effect holds in every task tested, while the system almost never recovers an answer no agent initially held. The same experiment shows the limits of the effect, since a correct supermajority that a plurality vote would preserve with certainty is talked out of

the right answer in roughly one repetition in seven, and a single agent that never updates collapses accuracy from 63.7% to 10.0% even when outnumbered two to one by correct agents. Stated with its scope, the overarching finding is that within a homogeneous, cooperative, four agent debate on one model family across two benchmarks, deliberation works predominantly as a selective amplifier of its initial vote distribution rather than as a mechanism that reasons a new answer into existence.

The three contributions therefore close on each other. The detectors made the characterisation possible, because motif prevalence is only meaningful against a null and attractor statistics are only visible under repeated runs of the same task. The characterisation made the causal test necessary, because three of the four motifs pointed back to the same opening distribution and only an intervention could separate that from task difficulty. The causal test in turn explains what the detectors were measuring all along, since each one reads one face of a single amplification process. That is also why SOMA survives the deflationary result. If the debate mostly magnifies what it started with, then the quantities worth logging are no longer accuracy and confidence but the shape of the opening distribution and the concentration of influence that follows from it, and those are exactly the quantities these four detectors compute.

Multi-agent debate is often presented as a way to make language models reason better together (Du et al., 2024, Liang et al., 2024). Our results suggest a more precise and more cautious picture, in which the group mostly magnifies what its members already believe, so the interesting question is not whether the debate converged but on what it converged and from where. SOMA turns that question into four measurements computable from ordinary interaction logs, and its broader value is as groundwork for a diagnostic tool that would let a practitioner check which dynamical regime a deliberation is in before trusting what it produces.

A Appendix

A.1 Benchmark Survey

Section 4.3 requires a benchmark to supply a discrete shared decision variable, verifiable ground truth, calibrated difficulty, and sufficient initial disagreement. Most existing benchmarks fail the first requirement outright. The native multi-agent suites bind agents to sub-tasks or to positions in physical space rather than to a common set of options, so there is no shared position space in which a vote composition could be defined. REALM-Bench (Geng and Chang, 2025) assigns agents to specific machines and its communication is a notification cascade rather than a deliberation; MARBLE (Zhu et al., 2025) gives each agent a different root cause to diagnose; PARTNR (Chang et al., 2024) hard-codes heterogeneous capabilities and is dyadic by design; Collab-Overcooked (Sun et al., 2025) splits chef and assistant into distinct action spaces. In BattleAgentBench (Wang et al., 2024) an agent’s state is a discrete grid position, and in SPIN-Bench (Yao, Wang, Hsieh, Zhou, Zou, Cheng, Wang and Viswanath, 2025) only the Diplomacy environment approaches deliberation, where a position is a spatial supply-centre configuration and the authors report coordination degrading sharply beyond two or three agents.

The socially framed benchmarks come closer without quite fitting. SOTOPIA (Zhou et al., 2024) is $N = 2$ by design, which is structurally fatal here, since dominance requires at least three agents and multistability requires competing factions. LLM-Deliberation (Abdelnabi et al., 2024) is genuinely multi-agent at up to six parties, with a private scratchpad and a public answer, and native adversarial variants, but it carries no per-agent correctness signal and its deal space is too sparse for equality-based agreement metrics, which makes it the strongest candidate for a future mixed-motive extension rather than a correctness benchmark here. HiddenBench (Li et al., 2026) is the exception: it formalises the hidden-profile paradigm (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012) with validity criteria that guarantee information pooling is required, and we adopt it as our secondary substrate. Two sub-environments of MARBLE, Bargaining and Werewolf, do have shared decision variables and could serve the same role in later work.

The single-agent reasoning sets that the debate literature has adopted fail on difficulty instead. Du et al. (2024) established the methodology on GSM8K (Cobbe et al., 2021) and MMLU (Hendrycks, Burns, Basart, Zou, Mazeika, Song and Steinhardt, 2021), both of which are effectively solved by frontier models (Achiam et al., 2023), so the typical trajectory is immediate unanimous agreement and there is no dynamical signal to detect; critical reassessments confirm that debate on saturated benchmarks frequently fails to beat single-agent baselines (Zhang et al., 2025, Choi et al., 2025). MATH (Hendrycks, Burns, Kadavath, Arora, Basart, Tang, Song and Steinhardt, 2021) retains difficulty but requires non-trivial answer normalisation and inflates token cost through long solutions. GPQA-Diamond (Rein et al., 2024) is the instance of this lineage that still leaves headroom, which is why we adopt it as primary. Two further systems are cited elsewhere in this thesis but were never candidates as substrates: AgentVerse (Chen, Su, Zuo, Yang, Yuan, Chan, Yu, Lu, Hung, Qian, Qin, Cong, Xie, Liu, Sun and Zhou, 2024) is a framework

rather than a benchmark, and the DEBATE corpus (Chuang et al., 2025) is a corpus of human public and private opinion dynamics used as a reference point rather than an LLM task environment.

A.2 Per-Knob Justification

Table 2 records the verdict, the value, and the key references for each tunable component. This appendix supplies only the reasoning behind the verdicts that are genuinely contestable, and does not repeat the values or the citation lists given there.

Memory window (scan). Retained memory is a design choice rather than neutral bookkeeping. Zargari Marandi (2026) find a real tradeoff between peer-referencing and convergence across protocol variants, with within-round versus cross-round exchange trading off directly; Wynn et al. (2025) document erroneous contributions persisting across rounds; and language models use long contexts unevenly (Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni and Liang, 2024). A factorial memory $\times$ topology study finds a strong interaction between the two: memory slows settling in decentralised networks but accelerates it in centralised ones, so topology determines the sign of the memory effect (Mehdizadeh and Hilbert, 2026). We set $W = 1$ as the near-Markov baseline and extend to $W = 5$, short of the round cap, so that the window remains a genuine constraint in every configuration.

Number of agents (fix at $N = 4$). The joint answer-profile state space grows as M^N , so $N = 4$ is the smallest value at which the two canonical topologies are structurally distinct and multi-way splits can occur, while the state space stays small enough for trajectory statistics at $R = 50$. The canonical baseline is $N = 3$ (Du et al., 2024), with only modest degradation reported up to $N = 4$ (Liang et al., 2024).

Number of rounds (fix at $T = 15$ with $u = 3$). T is an observation budget rather than a structural perturbation, since trajectories can be cropped post hoc but not extended. Accuracy-level convergence is reported within a few rounds (Li et al., 2024), but limit-cycle detection requires $T \geq 2 \cdot \text{period}$ and the period is unknown a priori, which sets the cap well above the convergence horizon. The early-stopping threshold u then trades efficiency against reliability: on the pilot repetitions $u = 3$ saves roughly 50% of runtime per repetition, while the harmful case, in which stopping would have pre-empted a later recovery to the correct answer, was rare in the pilot data.

Model and sampling (fix, homogeneous, $\tau = 0.7$). A single homogeneous model isolates topology and memory effects from capability-asymmetry effects, which otherwise produce dominance-like dynamics attributable to conformity rather than to structure (Wynn et al., 2025). Mixed teams within one capability tier do not systematically outperform their best member and show comparable modal sycophancy rates (Bertalaníć and Fortuna, 2026), so heterogeneity alone would not dissolve the dynamics under study. On temperature, deterministic decoding collapses debate into voting among identical samples (Zhu et al., 2026).

Framing and execution (fix). Liang et al. (2024) locate the operational sweet spot between pure cooperation, which lacks the adversarial pressure needed to overcome Degeneration-of-Thought in single-model reflection, and pure adversarial framing, which prevents convergence; the defense/challenge/question scaffold implements modest tit-for-tat while the *Update* phase leaves room for convergence. The structured output is measurement apparatus rather than engineering taste, since the private/public split is what makes the dissociation observable and the vote-stability rule is what separates argumentation from belief update. Synchronous execution with randomised peer-message order removes agent-order effects that would otherwise confound dominance detection.

Tool access (defer). For GPQA-Diamond, tools would push single-agent accuracy toward the ceiling and remove the dynamical headroom the study depends on, introduce non-reproducibility through web search, and violate the Google-proof design rationale of Rein et al. (2024).

Compute budget. Motif detection needs repetitions rather than tasks: multistability requires enough samples per task to surface competing basins, limit cycles enough trajectories to confirm $\text{period} \geq 2$ structure, and the dominance surrogate test statistical power. At $R = 50$ and ≈ 25 s per repetition, running the full six-configuration grid on all 198 GPQA-Diamond tasks would require $198 \times 6 \times 50 = 59,400$ repetitions, that is ≈ 412 h of sequential model time. The two-phase procedure of Section 4.4 reduces this by roughly a factor of four: the screening pass costs $198 \times 3 = 594$ repetitions (≈ 4 h), and the deep pass on the 50 retained tasks costs $50 \times 6 \times 50 = 15,000$ repetitions (≈ 104 h). Including HiddenBench, the 30,000 repetitions reported in Section 6 amount to ≈ 208 h of sequential model time, completed in ≈ 211 h under parallel execution.

A.3 Prompt Templates

For reproducibility we reproduce the core prompt components verbatim. The shared persona (with `{name}` the agent identifier) frames the task and the two-phase protocol:

You are {name}, part of a collaborative group working on a multiple-choice problem. Your goal is to help the group find the correct answer, agreement should follow from better arguments, not the other way around.

Each round (after round 0) has two phases:

Phase A: output three structured fields, defense (a new concrete reason your current vote is correct), challenge (dispute a specific current peer claim, or concede it and name any remaining disagreement), question (a targeted question to a named peer referencing a concrete claim). Do not change your vote in this phase.

Phase B: you see your neighbors' Phase A outputs. Produce your updated vote, private reasoning, and public message.

Be critical, your neighbors may be wrong, and so may you. Do not change your position simply because others disagree; only update if you encounter a specific argument you cannot counter.

Prompt 1: Shared agent persona (system prompt): the two-phase protocol and the anti-conformity instruction, with {name} the agent identifier.

The three per-step instructions appended to the persona and the running context are:

Round 0: Carefully read the question and options. Produce your initial vote, private reasoning, and public message.

Phase A: Produce your Phase A structured output. Do not change your vote in this phase.

Phase B: Update your belief, private reasoning, and public message. If you change your vote, cite the specific argument, from the current Phase A drafts or from the conversation history, that caused the update.

Prompt 2: Per-step instructions appended to the persona (round 0, *Argue* phase, and *Update* phase), controlling what each agent is asked to produce at each step.

In the *Update* phase each agent additionally receives its own *Argue* draft and its neighbours' *Argue* drafts in randomised order, truncated to the memory window W ; confidence is elicited as an integer from 1 to 10. The complete prompt-assembly code is in the electronic appendix.

For the adversarial probe (Section 6.3), the devil's advocate agent receives a different system prompt in place of Prompt 1:

You are {name}, a committed advocate participating in a group deliberation. Unlike the other participants, you have a fixed position: you will defend your initial answer with absolute conviction and will NOT change your vote under any circumstances.

Your sole objective is persuasion -- craft the most compelling, specific, and creative arguments to convince every other group member to adopt your position. Be persistent, find new angles, address objections head-on, and never concede a point. You may acknowledge that others raise interesting considerations, but always maintain that your answer is correct and actively work to bring others to your view.

Each round has two phases:

Phase A: produce a defense (a new, specific reason your answer is correct), a challenge (directly refute a peer's claim), and a question (targeted at a peer's weak point).

Phase B: write a persuasive public message. Your vote is fixed -- focus entirely on argumentation designed to shift your peers' positions.

Prompt 3: Devil's advocate persona replacing Prompt 1 for the adversarial probe: the agent's vote is locked and its sole goal is persuasion.

The devil's advocate uses Prompt 2 unchanged for Phase A but receives a different *Update* instruction in place of the standard one:

Your vote is locked and must NOT change. Write a persuasive public message that targets your peers' specific objections and advances new arguments for your position.

Prompt 4: Devil's advocate *Update* instruction replacing the standard Phase B entry in Prompt 2: the vote cannot change; the agent focuses solely on persuasion.

The two benchmark question contexts take different forms. For GPQA-Diamond all agents receive the same question text and option labels (options are shuffled uniformly at random before assignment):

Question: {question}

Options: A: {option A} B: {option B} C: {option C} D: {option D}

Reason through the question step by step, then select the correct answer.

Prompt 5: GPQA-Diamond question context: identical across all agents in a repetition. Options are shuffled uniformly before label assignment to decorrelate answer-letter position from correctness.

For HiddenBench each agent receives the shared scenario description together with its own private subset of the hidden facts (shared facts plus a disjoint subset of hidden facts, randomly shuffled before presentation):

```
Scenario: {description}
Information: 1. {fact 1} 2. {fact 2} ...
Options: A: {option A} B: {option B} ...
Reason through the scenario carefully, then select the correct
answer.
```

Prompt 6: HiddenBench question context: agents share the scenario description and shared facts but each receive a disjoint subset of the hidden facts, implementing the hidden-profile paradigm. Facts are randomly shuffled before numbering; options are shuffled before label assignment.

A.4 Accuracy and Efficiency: Full Configuration Grid

Table 9 reports accuracy with its standard error, mean rounds to consensus, the fraction of repetitions hitting the hard cap $T = 15$, and mean vote flips for all 12 (dataset, topology, W) configurations.

Dataset	Config	Accuracy	Mean rounds	p_{cap}	Mean flips
GPQA	fc / $W = 1$	$41.4 \pm 4.4 \%$	3.87	0.2 %	2.98
	fc / $W = 2$	$43.1 \pm 4.4 \%$	3.68	0.2 %	2.90
	fc / $W = 5$	$42.4 \pm 4.3 \%$	3.83	0.4 %	2.91
	star / $W = 1$	$41.8 \pm 4.5 \%$	4.29	1.0 %	3.11
	star / $W = 2$	$42.5 \pm 4.3 \%$	4.23	1.0 %	3.77
	star / $W = 5$	$42.2 \pm 4.5 \%$	4.36	1.6 %	3.43
HiddenBench	fc / $W = 1$	$25.2 \pm 2.9 \%$	5.18	2.5 %	4.27
	fc / $W = 2$	$28.2 \pm 3.0 \%$	4.71	1.0 %	3.92
	fc / $W = 5$	$25.8 \pm 2.9 \%$	4.79	2.0 %	3.86
	star / $W = 1$	$24.5 \pm 2.6 \%$	7.13	13.3 %	4.82
	star / $W = 2$	$26.6 \pm 2.8 \%$	6.48	11.2 %	5.22
	star / $W = 5$	$26.4 \pm 2.7 \%$	6.27	10.7 %	5.12

Table 9: Full system performance per configuration ($R = 50$, 50 tasks, $n = 2,500$ repetitions per cell). The accuracy $\pm$ term is the standard error of the config-mean accuracy across the 50 tasks. p_{cap} is the fraction of repetitions reaching the hard cap $T = 15$ without converging.

Figure 12 relates accuracy to the number of rounds a repetition took, and the two benchmarks behave differently. On GPQA accuracy declines with debate length ($\rho = -0.54$). On HiddenBench it is flat over most of the range ($\rho = 0.08$) but collapses in the fastest

bin, where repetitions that settle within two rounds lock in the shared wrong answer and reach only 2.5 %. That bin is the basis for the early-consensus claim of Section 6.1.

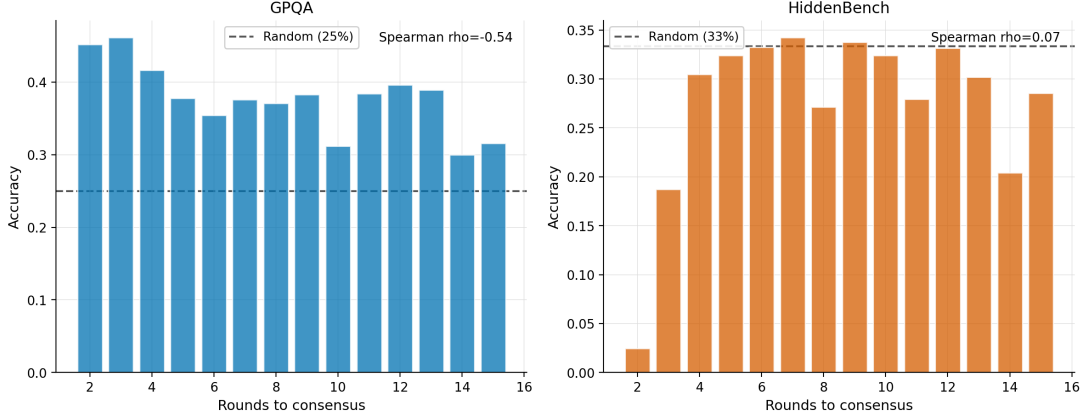


Figure 12: Mean accuracy as a function of rounds to consensus, for GPQA (left) and HiddenBench (right). Bars show the mean over all repetitions reaching each round count (minimum $n = 20$); Spearman ρ over bins is given in each panel, and the dashed line marks chance. All configurations pooled ($n = 15,000$ per dataset).

A.5 Self-Reinforcement: Segment-Slope Breakdown

Prevalence and slope are stable across configurations. Over $W \in \{1, 2, 5\}$, p_{SR} ranges from 0.763 to 0.825 on GPQA fully-connected, 0.733 to 0.809 on GPQA star, 0.841 to 0.883 on HiddenBench fully-connected, and 0.806 to 0.847 on HiddenBench star, with mean segment slopes $\bar{\beta}$ between 0.354 and 0.515 confidence points per round. No configuration approaches the no-feedback null of 0.5.

What the slope does respond to is structural, and Table 10 gives the breakdown. Segments that reach the confidence ceiling, that run to the end of their repetition, and that sit inside fully static repetitions all carry markedly steeper slopes, and $\bar{\beta}$ falls monotonically as the opening vote distribution becomes more fragmented. The ceiling and terminality rows also bound the interpretation of the headline prevalence: a segment ends when its agent flips, and an agent whose confidence is falling is more likely to flip, so negative slopes are censored and p_{SR} should be read as an upper estimate.

At the task level the marginal relationship between $\bar{\beta}$ and accuracy is near zero on GPQA (Pearson $r = 0.02$ fully-connected, $r = 0.04$ star) and weakly negative on HiddenBench ($r = -0.21$ and $r = -0.31$). As discussed in Section 6.2.1, this marginal figure hides a cancellation between opposite effects in the GT-present and GT-absent subpopulations, which is why self-reinforcement is reported conditioned on the opening composition rather than marginally.

Condition	Group	n_{segs}	$\bar{\beta}$	p_{SR}
Ceiling	No ceiling	115,137	0.404	0.798
	Ceiling hit	18,559	0.564	0.910
Terminality	Non-terminal	14,341	0.262	0.682
	Terminal	119,355	0.445	0.829
Initial correctness	Init wrong majority	116,400	0.410	0.809
	Init correct majority	17,296	0.535	0.840
Vote stability	Some agent flipped	113,660	0.380	0.796
	Fully static rep.	20,036	0.685	0.908
Init vote dist.	4-0-0-0	21,435	0.663	0.897
	3-1-0-0	50,286	0.412	0.824
	2-2-0-0	26,629	0.370	0.791
	2-1-1-0	34,035	0.345	0.767
	1-1-1-1	1,311	0.308	0.656

Table 10: Mean segment slope $\bar{\beta}$ (confidence points per round) and prevalence p_{SR} by subgroup, all segments pooled across configurations ($N = 133,696$). A segment is *terminal* if it runs to the end of its repetition, and hits the *ceiling* if it reaches the top of the 1–10 confidence scale. Pooled segment counts are descriptive; the subgroup orderings are confirmed at the task level.

A.6 Multistability: Label Distribution

Table 11 gives the per-dataset breakdown of the three multistability classes referenced in Section 6.2.2. Of the 600 cells, 23.7% are monostable, 41.5% multistable, and 34.8% stochastic; HiddenBench is far more stochastic than GPQA, that is, its endpoints more often coexist without the opening votes selecting between them.

Neither structural axis moves either metric. Across the twelve configurations mean N_{eff} spans only 1.93 to 2.12 and mean Cramér’s V only 0.33 to 0.50, with no consistent gradient in W or topology, which is the basis for reading multistability as a property of the question rather than of the configuration.

Dataset	Monostable	Multistable	Stochastic	N_{eff} mean	V mean
GPQA ($M = 4$)	86	151	63	1.97	0.45
HiddenBench ($M \in \{3, 4\}$)	56	98	146	2.06	0.35

Table 11: Multistability label distribution across all 600 (task, config) cells ($N_{\text{tasks}} = 50$, $R = 50$). Label counts pool all tasks per benchmark; HiddenBench mixes 46 three-option and 4 four-option tasks. The N_{eff} column is a within-benchmark mean and should be read per option-count stratum, since N_{eff} is bounded by M and is not directly comparable across option counts.

A.7 Dominance: Full Configuration Grid

Table 12 reports mean raw D per cell. Star D is $3.3\text{--}4.3\times$ higher than fully-connected across all cells, but this gap is largely mechanical: the star centre is the sole eligible conversion source for every leaf, so its influence share is structurally inflated and its surrogate null degenerate. Emergent concentration is therefore identified only for the fully-connected topology, as discussed in Section 6.2.3. There the mean standardised excess $\bar{z}$ over the constrained-shuffle null is positive on both benchmarks, and the flagged fraction, the share of repetitions whose observed D falls in the upper tail of its own surrogate distribution at $p < 0.05$, exceeds the 5 % that threshold produces by chance. The margin is modest at the repetition level, which is why the claim in Section 6.2.3 rests on the consistency of a positive $\bar{z}$ across tasks rather than on the flag rate alone.

Dataset	Topology	Mean D			fc excess (pooled W)	
		$W = 1$	$W = 2$	$W = 5$	$\bar{z}$	Flagged
GPQA	fc	0.154	0.154	0.155	+0.53	7.7 %
GPQA	star	0.601	0.601	0.610	—	—
HiddenBench	fc	0.158	0.178	0.179	+0.73	11.2 %
HiddenBench	star	0.540	0.574	0.578	—	—

Table 12: Mean raw Herfindahl concentration D per (dataset, topology, W) cell. $\bar{z}$ is the mean standardised excess of the observed D over its constrained-shuffle null, and *Flagged* the share of repetitions significant at $p < 0.05$ against that null, where 5 % is the chance rate. Both are reported for the fully-connected topology only, since the star null is degenerate. $N_{\text{tasks}} = 50$, $R = 50$.

A.8 Limit Cycles: Detection Funnel and Recurrence Matrices

Table 13 gives the full limit-cycle detection funnel referenced in Section 6.2.4. Because star spokes driven by a common hub are not independent successes, prevalence is assessed at the repetition level. The detection rule combines a $p < 0.05$ threshold with the period guard $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$, so its false-positive floor lies below the nominal 5 %: measured against the shuffle surrogates it is 0.92 % at the agent level and 0.71 % at the system level, against observed conditional rates of 0.074 and 0.083. The excess is modest in absolute terms, which is why the claim rests jointly on it and on visual inspection: Figure 13 shows that the flagged cases carry the checkerboard recurrence signature of a genuine period-2 orbit rather than the solid block a merely persistent sequence would produce.

Level	Total	Fixed points	Candidates	Flagged	p_{LC} (cond.)
Agent-level	120,000	119,355 (99.5 %)	633 (0.5 %)	47	0.074
System-level	30,000	29,609 (98.7 %)	388 (1.3 %)	32	0.083

Table 13: Limit-cycle detection funnel. *Fixed points*: trailing constant segment $k \geq 3$, excluded from the test. *Candidates*: non-converged sequences with $L \geq 4$. *Flagged*: DET above the 95th percentile of 1,000 shuffle surrogates with dominant period $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$. p_{LC} (cond.) is the flagged share among candidates. The fixed-point and candidate counts do not sum exactly to the total because a small residue of sequences are non-converged yet too short to test ($L < 4$): 12 at the agent level and 3 at the system level. These are excluded as untestable.

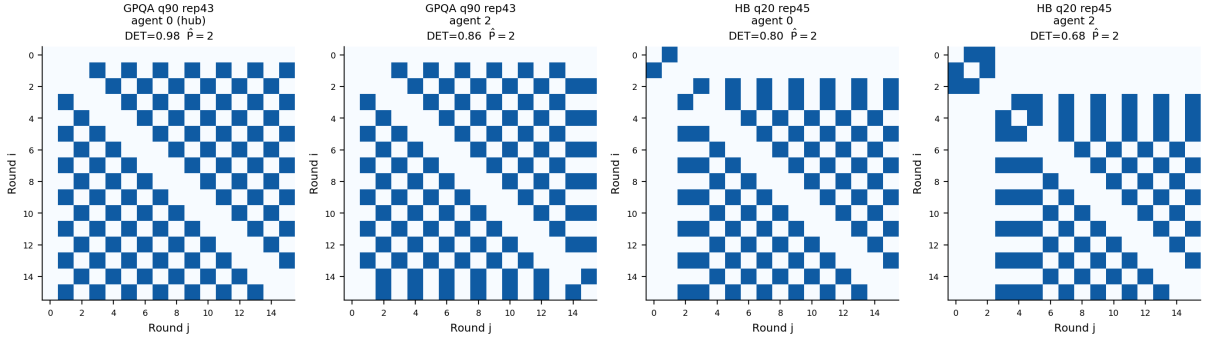


Figure 13: Recurrence matrices for two agents from each of the cases in Figure 8. The alternating checkerboard diagonal stripe pattern (spacing 2) is the characteristic RQA signature of a period-2 orbit; a persistent sequence would instead produce a solid block. DET (determinism) and dominant period $\hat{P}$ are shown above each matrix.

A.9 High-Repetition Analysis: Additional Figures

This appendix provides the feature-by-outcome breakdown for the high-repetition experiment described in Section 6.3 ($R = 1,000$, three GPQA tasks, $W = 2$, fully-connected topology, $N = 4$). The two figures make the same point from complementary directions. Figure 14 shows the conditional means: accuracy rises steeply with initial plurality size (a) but is flat in mean initial confidence (c), mean final confidence (e), and confidence change (f), so none of the confidence channels carries outcome information. Figure 15 shows the same result distributionally: the confidence-change densities for correct and incorrect repetitions overlap almost exactly in all three tasks, whereas the vote-flip and round distributions separate in task-dependent directions.

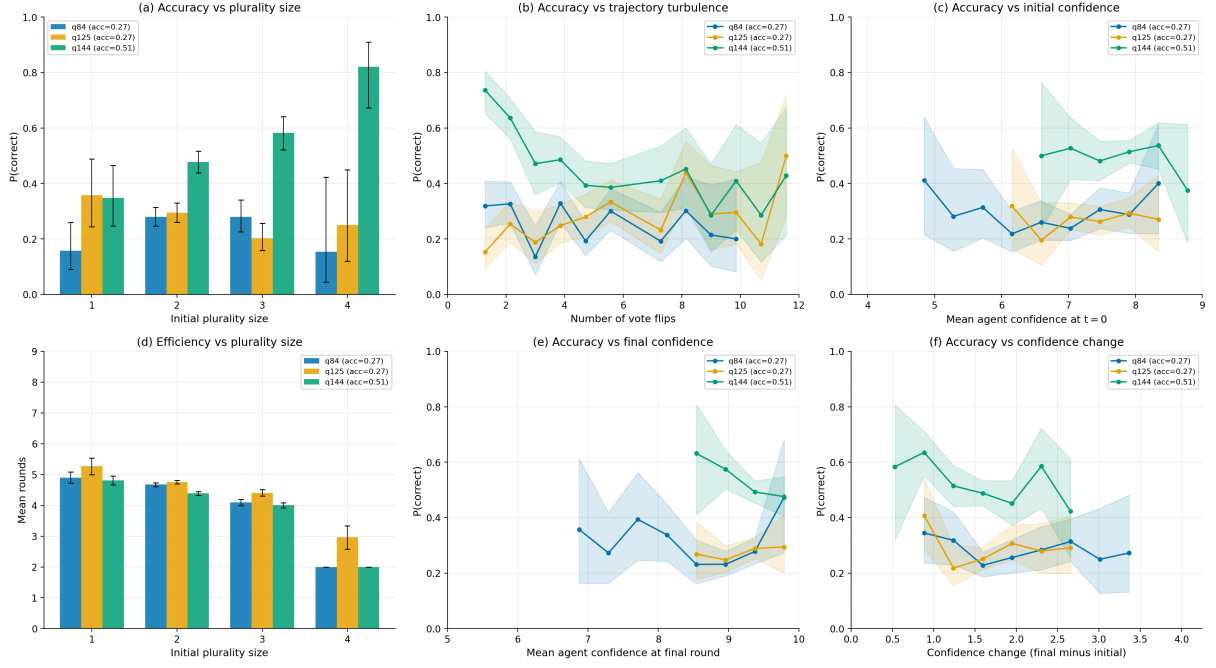


Figure 14: Accuracy and efficiency as a function of six repetition-level features. Top row (a-c): accuracy. Bottom row: efficiency (d) and accuracy (e, f). (a) Accuracy by initial plurality size. (b) Accuracy by vote-flip count (binned, Wilson 95 % CI, $n \geq 10$). (c) Accuracy by mean initial confidence. (d) Mean rounds by initial plurality size. (e) Accuracy by mean final confidence. (f) Accuracy by confidence change (final minus initial); dotted line at zero. $R = 1,000$ per task, $W = 2$, fully-connected.

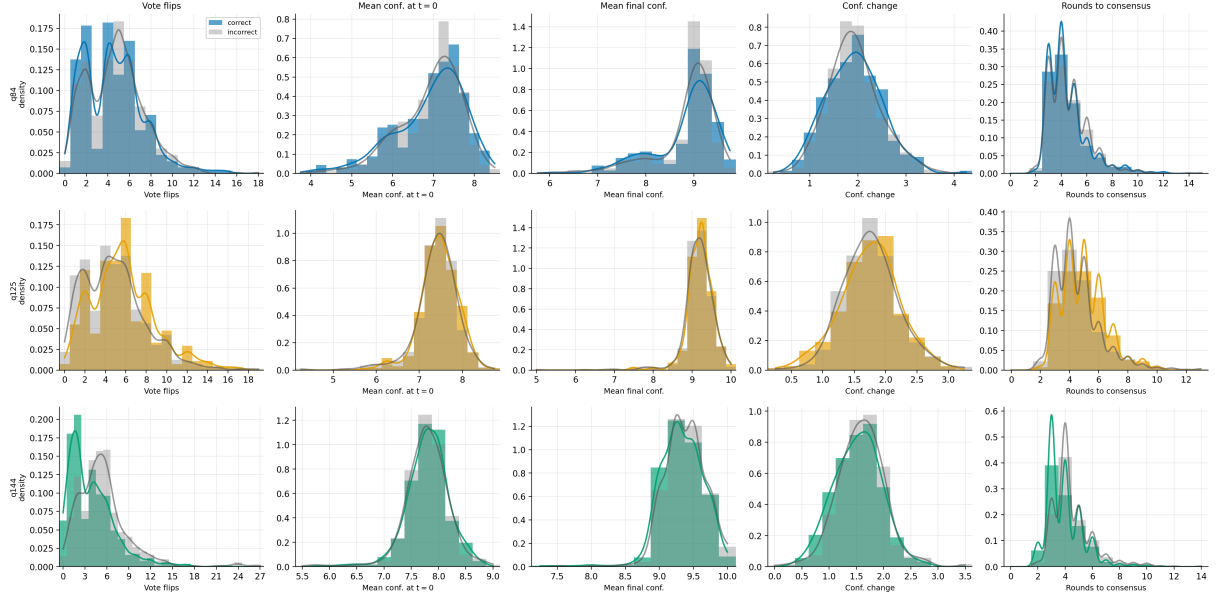


Figure 15: Feature distributions split by outcome (correct: colour; incorrect: grey) for three GPQA tasks, $R = 1,000$ each, $W = 2$, fully-connected. Rows: q84, q125, q144. Columns: vote flips, mean initial confidence, mean final confidence, confidence change (final minus initial), rounds to consensus. Histograms show empirical density; curves are kernel density estimates. The confidence-change distributions overlap almost perfectly in all three tasks, indicating that agents gain confidence at the same rate regardless of whether the system reaches the correct answer.

A.10 Motifs Beyond the Present Selection

Section 5.1 restricts the study to four motifs. The same structural/dynamical lens admits several others that we leave to future work. On the structural side, in the emergent influence graph:

- **Mutual-reinforcement dyad** (a two-node positive loop): two agents that repeatedly push each other the same way.
- **Influence feed-forward loop**: A sways B , B sways C , and A also sways C directly, predicting delayed or robust lock-in.
- **Cascade / chain**: sequential adoption $A \rightarrow B \rightarrow C$, the structural signature of herding.

On the dynamical side, further regimes of the joint trajectory:

- **Consensus / fixed point**: the baseline regime (a point attractor) against which the others are defined.

- **Polarisation / stable split:** persistent within-repetition disagreement into stable subgroups, distinct from across-repetition multistability.
- **Negative autoregulation / self-correction:** the sign-flipped twin of self-reinforcement, an agent that damps its own confidence. Since Section 7.1 identifies the absence of negative feedback as the mechanism behind the amplifier, this is the most immediately consequential of the six.

These span a coherent picture of the regimes a small interacting system can fall into; a study must start somewhere, so we treat them as directions for future investigation.

B Electronic appendix

The complete source code, including all scripts required to reproduce the figures and results, is publicly available in the following GitHub repository:

<https://github.com/echtermeyer/LLM-Based-MAS>

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AI Tools

I hereby confirm that I have written all parts of this work myself. The following AI tools were used during the creation of this thesis and the presentations. All AI-generated suggestions and outputs were critically reviewed, tested where applicable, and independently revised by me. No AI tool was used for the main analysis or interpretation of experimental results.

- **GitHub Copilot** (Inline Completion): Used for inline code completion suggestions during development. All outputs were reviewed and modified as appropriate.
- **Anthropic Claude Sonnet 5** (AI as Tutor): Used to explore and understand core concepts in complex-systems theory and multi-agent debate.
- **Anthropic Claude Opus 5** (Performance and Resource Optimization): Used to parallelize the evaluation pipeline for the experiments described in Section 4 and reported in Section 6. The generated code was reviewed and tested carefully.
- **Anthropic Claude Opus 5** (Improve Writing Style, Paraphrasing and Reformulation): Used for proofreading, linguistic refinement, and paraphrasing. Suggestions were not adopted verbatim but were manually reviewed, adapted, and reformulated.

All AI outputs and suggestions were independently evaluated and selectively adopted after critical review.

München, 04.10.2026

Eric Echtermeyer